

Independence and Minimality of the Phase Axioms

Marlon Hanks

Independent Theoretical Physics Research

May 11, 2026

Subject Areas: Foundations of Physics | Axiomatic Analysis | Unified Theory

Keywords: axiomatic independence, minimality, phase ontology, foundational axioms,
non-redundancy, Phase Theory

Cross-references: Papers 1, 3, 5, 6, 7, 8, 10 of the Phase Theory series

Abstract

We analyze the axiomatic foundation of Phase Theory with respect to independence, minimality, and non-redundancy. Each axiom introduced in Paper 1 is examined to determine whether it can be derived from the remaining axioms, weakened without loss of explanatory power, or eliminated entirely. We employ formal independence analysis adapted from mathematical logic and foundational physics, demonstrating that the six axioms of Phase Theory form a minimal generating set: removing or weakening any single axiom either reintroduces forbidden ontological primitives, permits physically inadmissible configurations, or renders the theory non-predictive with respect to at least one downstream result. We further demonstrate that no axiom is a logical consequence of any subset of the remaining axioms, establishing the full logical independence of the set. The paper includes a formal minimality theorem, a pairwise independence analysis, a structural classification of the axioms by functional role, a dependency graph connecting each axiom to the physical

structures it underwrites, comparisons with competing foundational frameworks, and a systematic treatment of objections. Phase Theory's axiomatic commitments are shown to be irreducible within its stated scope.

Contents

1. Introduction: Why Axiom Independence Matters
2. Scope and Methodology
3. Recap of the Phase Theory Axioms
4. Formal Framework for Independence Analysis
5. Axiom I: Phase Primacy — Statement and Conceptual Context
6. Axiom I: Independence Proof
7. Axiom I: Necessity Analysis
8. Axiom II: Phase Continuity — Statement and Conceptual Context
9. Axiom II: Independence Proof
10. Axiom II: Necessity Analysis — Continuous vs. Discrete Phase
11. Axiom III: Phase Admissibility — Statement and Conceptual Context
12. Axiom III: Independence Proof
13. Axiom III: Necessity Analysis — The Selection Principle
14. Axiom IV: Phase Evolution — Statement and Conceptual Context
15. Axiom IV: Independence Proof
16. Axiom IV: Necessity Analysis — Dynamics, Causality, and Time
17. Axiom V: Observability — Statement and Conceptual Context
18. Axiom V: Independence Proof
19. Axiom V: Necessity Analysis — Empirical Contact and Measurement Consistency
20. Axiom VI: Closure — Statement and Conceptual Context

21. Axiom VI: Independence Proof
 22. Axiom VI: Necessity Analysis — Ontological Stability and Reduction Guarantees
 23. The Minimality Theorem: Formal Statement
 24. Proof of the Minimality Theorem
 25. Non-Redundancy of the Axiom Set
 26. Pairwise Logical Independence: A Complete Analysis
 27. Structural Roles: A Classification of the Six Axioms
 28. Interdependence Structure and Dependency Graph
 29. Comparison with Other Foundational Theories
 30. The Ontological Economy Principle
 31. Axiom Strengthening: What Happens If We Strengthen Rather Than Weaken?
 32. Axiom Decomposition: Can Any Axiom Be Split?
 33. Objections and Replies
 34. The Relationship Between Axioms VI and I
 35. Implications for Falsifiability
 36. Connections to Gödel Completeness and Consistency
 37. Implications for Theory Extension
 38. The Role of Each Axiom in Downstream Papers
 39. Future Directions: Can the Axiom Count Be Reduced Below Six?
 40. Conclusion
- References
-

1. Introduction: Why Axiom Independence Matters

The history of foundational physics is, in substantial measure, a history of unrecognized redundancy. The axiom sets of classical mechanics, quantum mechanics, and general relativity have each, at various points in their

development, contained postulates that were either derivable from others, stronger than necessary, or concealing hidden assumptions whose independent status was not initially appreciated. Newton's first law of motion — the principle of inertia — has been shown in many modern formalizations to be a special case of Newton's second law applied to the limiting case of zero net force, calling into question whether it constitutes an independent axiom or a corollary [11]. The von Neumann axioms for quantum mechanics have been subjected to sustained scrutiny: the Born rule's independence from the remaining postulates remains contested [14], and the collapse postulate is widely regarded as either derivable from decoherence or independently superfluous [18]. General relativity's Bianchi identities impose algebraic dependencies among the components of the Einstein field equations, creating an internal redundancy that, while not fatal, complicates the identification of genuinely independent physical content [19]. These redundancies are not merely aesthetic blemishes. A foundational theory that contains redundant axioms suffers from a precise diagnostic deficit: when the theory makes a prediction that fails, one cannot determine which axiom is responsible. The redundancy distributes explanatory and predictive responsibility across axioms in ways that cannot be cleanly separated, making the theory harder to refine, extend, or falsify in a targeted manner.

Phase Theory, introduced in Paper 1 of this series [1], occupies an unusual position in this landscape. Unlike theories that acknowledge multiple ontological primitives, Phase Theory makes a maximal ontological reduction claim: there exists exactly one primitive entity, the global phase substrate Φ , from which all physical structure — matter, light, spacetime, probability, thermodynamics, and measurement — is derived. This claim is encoded in six axioms. The stakes of axiom redundancy are correspondingly elevated. If any axiom of Phase Theory can be derived from the remaining five, then the theory's axiomatic presentation is not the lean, irreducible foundation it claims to be, but a somewhat prolix restatement containing hidden

dependencies. More seriously: if any axiom is secretly a consequence of others, then the theory may be making a stronger commitment than necessary to reach its conclusions, which would mean that a simpler — and philosophically more defensible — foundation might exist. Conversely, if any axiom is too weak (can be further weakened without losing predictions), then Phase Theory is not operating at its proper level of generality and risks admitting configurations that were meant to be excluded. Both possibilities are formally damaging for a theory that presents itself as the minimally sufficient foundation for a complete description of physical reality.

This paper addresses a single, precisely stated question: Is the set $S = \{\text{Axiom I, Axiom II, Axiom III, Axiom IV, Axiom V, Axiom VI}\}$ the smallest possible axiom set that still generates the full predictive and explanatory framework of Phase Theory? We call this the **minimality question**. Answering it requires formal analysis of three distinct failure modes that an axiom set can exhibit. The first is **redundancy**: an axiom A_k is redundant if it can be logically derived from the remaining axioms in $S \setminus \{A_k\}$. A redundant axiom adds no independent content and should be reclassified as a theorem. The second failure mode is **excess strength**: an axiom A_k is unnecessarily strong if it can be replaced by a strictly weaker assertion $A_{k'}$ — one that is logically implied by A_k but does not imply A_k — and the weakened set $\{A_{k'}\} \cup (S \setminus \{A_k\})$ still recovers all core predictions. In this case, A_k over-commits the theory, forbidding configurations that the theory's predictions do not actually require to be forbidden. The third failure mode is **incompleteness**: the axiom set implicitly relies on assumptions not made explicit, meaning that the apparent derivations of downstream results actually require additional premises that have not been stated. We show that Phase Theory's axiom set exhibits none of these failure modes.

The formal machinery employed in this paper is adapted from two sources. First, the model-theoretic tradition of axiomatic analysis in mathematics,

specifically the technique of constructing explicit models that satisfy all but one axiom of a given set, thereby proving the logical independence of that axiom [8, 9, 10]. This technique was pioneered in the context of geometry (Euclid's parallel postulate), refined in the context of set theory (the independence of the Axiom of Choice and the Continuum Hypothesis from ZFC) [24], and has been applied, less formally but fruitfully, to the axiomatics of physical theories. Second, the model-theoretic semantics of physical theories following the tradition of suppes and van Fraassen, in which a physical theory is identified with a class of models and a theory's empirical predictions are identified with the common predictions of all models in the class. Axiomatic independence in this framework means that removing an axiom strictly enlarges the class of models, admitting models that violate at least one core prediction. The paper proceeds axiom by axiom through Sections 5–22, constructing explicit distinguishing models, before assembling the results into the formal minimality theorem (Section 23) and its proof (Section 24). The remaining sections develop implications and address objections.

We emphasize at the outset that this paper holds Phase Theory to the strictest standard of axiomatic discipline precisely because its claims are the most ambitious. A theory that posits a single primitive and claims to derive all of physics from six axioms must demonstrate that those six axioms are exactly the right number — neither one too many nor one too few. The positive conclusion of this paper is that Phase Theory meets this standard. The six axioms form a minimal, non-redundant, appropriately calibrated foundation. This result strengthens the foundational case for Phase Theory and sharpens the theory's empirical profile by establishing a direct correspondence between each axiom and the physical predictions it underwrites.

2. Scope and Methodology

The methodology of this paper requires careful preliminary articulation, since the notion of "axiomatic independence" admits several distinct interpretations that must be clearly distinguished. We recognize three principal senses in which an axiom may be called independent, and we apply all three throughout our analysis. The first sense is **logical independence**: an axiom A_k is logically independent of a set T of axioms if A_k is neither provable from T nor refutable from T in the relevant deductive system. This is the strongest and most purely formal sense of independence. To establish logical independence of A_k from $S \setminus \{A_k\}$, one constructs two models: a model M^+ satisfying all axioms in S (showing that A_k is consistent with the remaining axioms) and a model M^- satisfying all axioms in $S \setminus \{A_k\}$ but violating A_k (showing that A_k does not follow from the remaining axioms). In this paper, the existence of M^+ is established by Phase Theory itself; the models M^- are constructed explicitly in Sections 6, 9, 12, 15, 18, and 21.

The second sense is **physical necessity**: an axiom A_k is physically necessary if removing it causes the theory to lose at least one core physical prediction — that is, if the weakened theory $S \setminus \{A_k\}$ admits models that make predictions contradicting or failing to recover results established in the downstream papers of the Phase Theory series. Physical necessity is a model-theoretic rather than a purely syntactic condition: it requires identifying what physical configurations become possible in the absence of A_k and whether those configurations are empirically admissible. This sense of independence is weaker than logical independence in the sense that a logically independent axiom may or may not be physically necessary; however, in Phase Theory's case, we show that the two conditions coincide for each of the six axioms.

The third sense, applied in Section 31, is **calibration**: an axiom A_k is correctly calibrated if neither weakening it nor strengthening it improves the theory's economy or generality. An axiom that is weaker than necessary

admits configurations the theory should exclude, while an axiom that is stronger than necessary excludes configurations the theory requires. Calibration analysis therefore examines both directions of potential modification for each axiom. The notion of **minimal axiom set** employed in this paper follows the tradition established by Kolmogorov in his axiomatization of probability theory [9], by Zermelo and Fraenkel in their axiomatization of set theory [10], and anticipated by Newton's own stated methodological principle — *hypotheses non fingo* ("I feign no hypotheses") — the injunction against postulating anything that is not required by the phenomena [11]. In each of these foundational traditions, minimality means that the axiom set is the smallest subset of propositions from which the entire framework can be derived, with no axiom serving merely as a convenient summary of consequences already entailed by the others.

We further distinguish between three types of potential reduction of an axiom set that are often conflated in informal discussion. **Axiom elimination** means removing an axiom entirely, which requires showing that all of the axiom's consequences are already derivable from the remaining axioms. **Axiom weakening** means replacing an axiom with a logically weaker statement that preserves the relevant consequences while reducing ontological or mathematical commitment. **Axiom decomposition** means identifying that an axiom is a conjunction of two logically independent statements, each of which should be listed separately; this increases the apparent axiom count but reflects more accurately the structure of the commitments being made. Sections 6–22 address axiom elimination; Section 31 addresses axiom strengthening (the complement of weakening) and, implicitly, weakening through the calibration analysis; Section 32 addresses axiom decomposition. The paper considers all three reduction types and shows that none is applicable to Phase Theory's axiom set.

A methodological note regarding the difference between **syntactic independence** and **semantic independence** is in order. Syntactic independence is a property of derivations within a formal deductive system: A_k is syntactically independent of T if there is no formal proof of A_k from T using the inference rules of the system. Semantic independence is a property of models: A_k is semantically independent of T if there is a model satisfying T but falsifying A_k . By Gödel's completeness theorem (for first-order logic), these two notions coincide: A_k is syntactically independent of T if and only if it is semantically independent [23]. The independence arguments of this paper are primarily semantic — they proceed by constructing distinguishing models — and are therefore fully rigorous within the scope of the physical framework. A fully syntactic treatment would require formalizing Phase Theory within a specific deductive system (for example, first-order logic over the language of differential geometry), which is a valuable but separate undertaking deferred to a companion mathematical paper.

3. Recap of the Phase Theory Axioms

For precision and self-containedness, we restate the six axioms of Phase Theory exactly as formulated in Paper 1 [1], providing for each axiom its formal role and the downstream papers that depend most heavily upon it. The axioms are understood as a jointly necessary and sufficient foundation for the theory: together they generate the entire Phase Theory framework; individually, each is required for a specific portion of the framework's predictive content.

Axiom I — Phase Primacy: There exists a global phase substrate Φ such that every physical entity, observable, and dynamical process is an emergent structure of Φ . No physical entity exists independently of Φ . This axiom establishes the ontological inventory of Phase Theory. It asserts both the existence of Φ and the non-existence of any entity not grounded in Φ .

Formally, it functions as an existence postulate coupled with a uniqueness constraint on the primitive ontology of the theory. Papers 3, 5, 6, 7, 8, and 10 all depend on this axiom as their foundational substrate.

Axiom II — Phase Continuity: The phase substrate Φ is a continuous and differentiable field except at a set of measure-zero singularities classified as admissible by Axiom III. Arbitrary discontinuities are forbidden. This axiom establishes the regularity class of Φ . It functions as a mathematical characterization axiom: given that Φ exists (Axiom I), Axiom II specifies the class of mathematical objects from which Φ is drawn, namely C^1 fields with a specified singularity structure. Papers 3 and 10 depend most directly on this axiom, since the emergence of smooth spacetime geometry requires continuous phase gradients, and the emergence of discreteness as a non-fundamental structure requires that the substrate itself be continuous.

Axiom III — Phase Admissibility: Not all phase configurations are physically realizable. Realizable configurations are those that extremize a global functional $A[\Phi]$ subject to topological and boundary constraints. The functional $A[\Phi]$ encodes the selection principle of the theory. This axiom introduces the dynamical law of Phase Theory in the form of a variational principle. Unlike traditional field-theoretic action principles, $A[\Phi]$ is not defined on a spacetime background but acts directly on the space of phase configurations. Papers 5, 7, and 8 depend most heavily on this axiom: the Born rule derivation requires that probabilities arise from the measure induced by admissibility; the finite particle spectrum requires that only finitely many topological classes are selected; the thermodynamic arrow of time requires a preferred equilibrium configuration determined by $A[\Phi]$.

Axiom IV — Phase Evolution: Physical evolution is the ordered sequence of admissible transformations of Φ . Time is not a primitive; the causal ordering of phase configurations constitutes emergent temporal structure. This axiom introduces dynamics into Phase Theory without presupposing temporal

structure as a primitive. It distinguishes Phase Theory from static variational theories by asserting that admissible configurations are not merely a set but an ordered sequence. Papers 4 and 8 depend most directly on this axiom, as cosmological evolution and the thermodynamic arrow of time both require a notion of ordered change.

Axiom V — Observability: All physical observables correspond to gauge-invariant invariants, extrema, or topological features of Φ . No quantity not expressible as such a feature of Φ is a physical observable. This axiom establishes the bridge between the abstract phase substrate and the experimental domain. It plays the role of Dirac's physical observable condition in the context of constrained Hamiltonian systems. Paper 6 (measurement theory) depends exclusively on this axiom for its consistency; Paper 5 and Paper 7 also rely on it in their identification of phase-theoretic quantities with observable physical magnitudes.

Axiom VI — Closure: No additional ontological primitives beyond Φ may be introduced at any stage of theory development. Fields, particles, spacetime, and observers are emergent structures of Φ , not co-primitives. This axiom is distinctive in character: it constrains the theory's future development rather than characterizing the present content of Φ . It is a second-order or meta-axiom that constitutionalizes ontological minimalism as a permanent constraint on Phase Theory's research program. Paper 7's prediction of a finite particle spectrum depends on Closure: if additional primitives could be introduced, additional particle species could be postulated rather than derived. All downstream papers rely on Closure implicitly in using derivation rather than postulation as their mode of theoretical development.

4. Formal Framework for Independence Analysis

We now establish the formal framework within which the independence analyses of Sections 5-22 are conducted. Let the axiom set be $S = \{A_I, A_{II}, A_{III}, A_{IV}, A_V, A_{VI}\}$, where each A_k denotes the corresponding axiom of Phase Theory. A **model** M for a subset $T \subseteq S$ is a mathematical structure (or physical theory) that satisfies every axiom in T . We say that M *satisfies* an axiom A_k if M is consistent with the statement of A_k under a natural interpretation of the axiom's terms within M 's framework. A model M *violates* A_k if M is consistent with the negation of A_k , or equivalently if A_k makes a non-trivial additional restriction on M that is not present in M 's structure.

The notion of **logical independence** is made precise as follows: an axiom $A_k \in S$ is logically independent of $S \setminus \{A_k\}$ if and only if there exists a model M_k such that M_k satisfies every axiom in $S \setminus \{A_k\}$ but M_k violates A_k . Such a model is called a **distinguishing model** or **independence witness** for A_k . The existence of M_k proves that A_k is not a logical consequence of $S \setminus \{A_k\}$: if it were a consequence, then every model of $S \setminus \{A_k\}$ would be a model of A_k , but M_k is a model of $S \setminus \{A_k\}$ that is not a model of A_k . This strategy is the direct analogue of proving the independence of Euclid's parallel postulate by constructing models of the other postulates (e.g., spherical geometry or hyperbolic geometry) in which the parallel postulate fails.

The notion of **minimality** is made precise as follows: let $P = \{P_3, P_5, P_6, P_7, P_8, P_{10}\}$ be the set of core physical predictions of Phase Theory, indexed by the downstream papers in which they are established. The set S is **minimal with respect to P** if for every $A_k \in S$, the reduced axiom set $S_k = S \setminus \{A_k\}$ fails to recover at least one prediction in P . Failure to recover a prediction P_j means that there exists a model M of S_k in which P_j does not

hold; in other words, S_k is consistent with the negation of P_j . This is strictly stronger than logical independence: it asserts not just that A_k is not derivable from $S \setminus \{A_k\}$, but that the loss of A_k has specific, identifiable physical consequences. The core physical predictions are formally characterized as follows:

- P_3 : The phase substrate Φ admits structures that behave as discrete entities at emergent scales, even though Φ itself is continuous.
Discreteness is not fundamental but derived from the admissibility structure of Φ .
- P_5 : The Born rule for quantum mechanical probabilities is a theorem derivable from the Phase Theory axioms, not an independent postulate. Probabilities arise from the measure induced by $A[\Phi]$ on the space of admissible configurations.
- P_6 : Measurement theory is consistent without a collapse postulate. The apparent reduction of a quantum state upon measurement is derivable from the phase-theoretic description of the measurement interaction, requiring no additional axiom.
- P_7 : The particle spectrum of Phase Theory is finite and bounded, containing approximately 25 ± 10 distinct species, each corresponding to a topological class of admissible singularities in Φ .
- P_8 : The thermodynamic arrow of time and the approach to equilibrium are derivable from the ordering of admissible phase transformations and the structure of $A[\Phi]$, without appeal to ad hoc initial conditions.
- P_{10} : The pseudo-Riemannian metric structure of observed spacetime emerges from smooth phase gradients $\nabla\Phi$, with gravitational effects corresponding to phase curvature.

Three key theorems are to be established in this paper. The first is **Theorem T1 (Logical Independence)**: Each axiom $A_k \in S$ is logically independent of $S \setminus \{A_k\}$. The second is **Theorem T2 (Minimality)**: The set S is minimal with respect to the core predictions P . The third is **Theorem T3 (Non-Redundancy)**: No axiom in S is a logical consequence of any proper subset of S . T3 is a strengthening of T1: T1 says A_k is independent of $S \setminus \{A_k\}$, while T3 says A_k is independent of every proper subset of S that does not include it. Since the only proper subsets of S not containing A_k are subsets of $S \setminus \{A_k\}$, T3 follows immediately from T1 by monotonicity of logical consequence: if A_k is not derivable from $S \setminus \{A_k\}$, it is a fortiori not derivable from any subset thereof. The formal proof of T3 is therefore a corollary of T1, established in Section 25. The more substantive independent argument given there provides additional confirmation through direct pairwise analysis.

We define **physical admissibility** formally as follows: a phase configuration Φ_0 is physically admissible if and only if (i) Φ_0 satisfies the regularity conditions of Axiom II on its domain $D \subseteq \text{Phase Configuration Space}$; (ii) $\delta A[\Phi_0] = 0$ (Eq. 1), where δ denotes functional variation with respect to perturbations of Φ_0 satisfying the boundary conditions specified in Axiom III; and (iii) the topological invariants of Φ_0 lie within the admissible class determined by the boundary constraints. A physically admissible **transformation** $T: \Phi \rightarrow \Phi'$ is a mapping from one admissible configuration to another that preserves all topological invariants of Φ except at finitely many localized transition events. An **admissible sequence** is a sequence $(\Phi_0, \Phi_1, \Phi_2, \dots)$ of admissible configurations connected by admissible transformations, ordered by the causal relation defined in Axiom IV. The full Phase Theory framework is the mathematical study of such admissible sequences and their invariant features.

5. Axiom I: Phase Primacy — Statement and Conceptual Context

Axiom I — Phase Primacy — is the axiom from which all others derive their significance. It asserts that the global phase substrate Φ is the unique fundamental entity: everything physical is either Φ itself or a structural feature of Φ . No object, field, spacetime point, probability distribution, or observer exists as an independent entity; each is a mode of organization of Φ . This is a claim of a specific and radical type: it is an ontological monism — the thesis that reality is ultimately one kind of thing — articulated with sufficient precision to generate physical predictions. Ontological monism has a long philosophical history, but its appearance in the context of foundational physics as a precisely constrained and falsifiable commitment is distinctive to Phase Theory.

To appreciate the strength of Axiom I, it is essential to contrast it with the less ambitious reduction programs that preceded it. Mach's principle — the thesis that inertia is determined by the global distribution of matter — sought to reduce inertia to relational facts about matter, eliminating absolute space as an independent primitive [20]. It succeeded only partially: Einstein incorporated Machian ideas into general relativity but was unable to fully eliminate background structure. Wheeler's "it from bit" program [21] proposed that information, rather than matter or energy, is the fundamental physical primitive; but this program, while philosophically provocative, did not generate a quantitative physical theory with specific falsifiable predictions. Penrose's twistor program [15] sought to replace spacetime with a complex projective space of twistors as the fundamental entity, generating spacetime as a secondary structure — but retained the metric structure of the twistor space as a background, introducing a primitive of its own. In each case, ontological reduction was partial: one primitive was absorbed at the cost of retaining or introducing others.

Phase Theory's Axiom I is structurally stronger than any of these precedents: it does not reduce one primitive to another, but asserts a single primitive from which all others emerge. Crucially, Φ is *not* a field on spacetime — which would presuppose spacetime as a background and reduce Axiom I to a claim about field theory rather than ontological primacy. Φ is the substrate from which spacetime itself is constructed in Paper 10. This distinguishes Phase Theory from all field-theoretic approaches, including the Standard Model, where fields are defined on a Minkowski or curved background that is specified independently. It also distinguishes Phase Theory from string theory and loop quantum gravity: string theory defines strings on a background manifold, retaining manifold topology as a primitive; loop quantum gravity defines spin networks and spin foams that encode geometry, but the combinatorial structure of those networks is itself a kind of background. Phase Theory admits no background structure of any kind at the fundamental level.

The technical content of Axiom I is richer than it might appear. "Every physical entity is an emergent structure of Φ " is a claim about what exists. "No physical entity exists independently of Φ " is a claim about what does not exist — it is a negative existential, asserting the absence of Φ -independent entities. Together they constitute a biconditional: X is a physical entity if and only if X is an emergent structure of Φ . This biconditional is the foundation of the theory's explanatory unity. Every derivation in every downstream paper is, implicitly, a demonstration that some feature of the physical world — a particle, a spacetime point, a probability, a thermodynamic equilibrium — is expressible as a structural feature of Φ . Axiom I guarantees that this program of derivation is both possible (since everything is a feature of Φ) and necessary (since nothing exists independently of Φ , derivability from Φ is the only form of physical explanation available within the theory).

6. Axiom I: Independence Proof

We now prove that Axiom I is logically independent of $\{A_{II}, A_{III}, A_{IV}, A_V, A_{VI}\}$. To do so, we construct an explicit independence witness: a model M_1 that satisfies all five of Axioms II–VI but violates Axiom I.

Let M_1 be a **dualistic field theory** defined as follows. M_1 posits two ontological primitives: a global phase substrate Φ (as in Phase Theory) and an independent pseudo-Riemannian spacetime manifold $\mathcal{M}$ with a fixed background metric $g_{\mu\nu}$. The field Φ is defined as a map $\Phi: \mathcal{M} \rightarrow \mathbb{C}$, assigning a complex phase value to each point of $\mathcal{M}$. We verify that M_1 satisfies Axioms II–VI. Axiom II (Continuity) is satisfied: Φ is continuous and differentiable on $\mathcal{M}$ except at a measure-zero set of admissible singularities (particle-like defects in the phase field). Axiom III (Admissibility) is satisfied: we define an admissibility functional $A_1[\Phi] = \int_{\mathcal{M}} L(\Phi, \nabla\Phi) d^4x$ (Eq. 2), where d^4x is the volume element of $\mathcal{M}$ and L is a Lagrangian density; configurations realizing the extrema of A_1 are admissible. Axiom IV (Evolution) is satisfied: the Euler-Lagrange equations derived from A_1 define an ordered sequence of admissible phase configurations, constituting a well-defined dynamical evolution. Axiom V (Observability) is satisfied: we restrict observables to gauge-invariant invariants, extrema, and topological features of Φ . Axiom VI (Closure) is satisfied as formulated relative to the axiom set: M_1 introduces no *additional* primitives beyond those already established in its foundation — Φ and $\mathcal{M}$ are both present from the outset, and no further primitives are added during theory development.

Now we verify that M_1 violates Axiom I. In M_1 , the spacetime manifold $\mathcal{M}$ exists independently of Φ : it is a background structure on which Φ is defined, not an emergent structure derived from Φ . In particular, the metric $g_{\mu\nu}$ — which encodes the causal and geometric structure of $\mathcal{M}$ — is a primitive of M_1 , not a feature of Φ . As a consequence, M_1 admits observables that are not expressible as features of Φ alone: for example, the proper time between

two spacetime events, the geodesic distance, or the Riemann curvature tensor are all definable directly from $g_{\mu\nu}$ without reference to Φ . These are genuine physical observables in M_1 — they have direct experimental correlates — and they violate Axiom V only if Axiom I is already in force. Since we are testing the independence of Axiom I, we may not invoke Axiom I to conclude that such metric observables are forbidden. Therefore M_1 is a consistent model of $\{A_{II}, A_{III}, A_{IV}, A_V, A_{VI}\}$ that violates A_I , establishing the logical independence of Axiom I.

The physical significance of M_1 is not merely technical. M_1 corresponds precisely to **semiclassical quantum gravity on a fixed background** — specifically, quantum field theory on a curved spacetime background in the Bunch-Davies or analogous state. This is a well-defined, internally consistent, and empirically successful physical framework used in cosmology (Hawking radiation, inflationary perturbation theory) and high-energy physics (QFT in curved spacetime). The independence of Axiom I is therefore not a formal artifact: it reflects a genuine physical alternative to Phase Theory, one that retains phase structure (in the form of quantum fields with phase degrees of freedom) but treats spacetime as a primitive. The existence of this alternative — theoretically coherent and empirically viable in its domain — confirms that Axiom I's assertion of full phase primacy is a substantive additional commitment, not entailed by the mathematical machinery of the remaining axioms.

This completes the proof of the logical independence of Axiom I from $\{A_{II}, A_{III}, A_{IV}, A_V, A_{VI}\}$. ■

7. Axiom I: Necessity Analysis

We now show that removing Axiom I from Phase Theory causes irreversible predictive collapse — specifically, that the reduced axiom set $S_I = \{A_{II}, A_{III}, A_{IV}, A_V, A_{VI}\}$ fails to recover at least one core physical prediction in

P. The distinguishing model M_1 constructed in Section 6 is the vehicle for this analysis.

In M_1 (dualistic theory with independent spacetime manifold $\mathcal{M}$), the derivation of spacetime geometry from phase structure (Paper 10's prediction P_10) fails. In Paper 10, the emergent spacetime metric is expressed as: $g_{\mu\nu} \propto \partial_\mu \Phi \cdot \partial_\nu \Phi^*$ (Eq. 3), where the metric arises from the phase gradients. This derivation requires that spacetime have no prior metric: the geometry is constructed from Φ , not imposed upon Φ . In M_1 , spacetime comes equipped with a pre-existing metric $g_{\mu\nu}^{\{(0)\}}$, and Eq. 3, if imposed, would produce a second "phase-derived" metric that is generically distinct from $g_{\mu\nu}^{\{(0)\}}$. The two metrics are not in general compatible, producing a self-inconsistency unless one is discarded. The model M_1 therefore either abandons the derivation of spacetime from phase (in which case P_10 fails) or acknowledges the contradiction (in which case M_1 is internally inconsistent for Phase Theory's purposes). Either way, P_10 is not recoverable from S_I alone.

In the absence of Axiom I, the reduced theory also admits the model M_1 with hidden variable supplements. Since $\mathcal{M}$ is an independent primitive in M_1 , it can support additional structure — for example, a hidden variable field $\lambda: \mathcal{M} \rightarrow \mathbb{R}$ defined independently of Φ . Such hidden variables are not excluded by any of Axioms II–VI in isolation, since those axioms constrain Φ but do not assert that Φ is the sole determiner of physical outcomes. The probability emergence result of Paper 5 (prediction P_5) derives the Born rule from the measure induced by $A[\Phi]$ on phase configuration space, under the assumption that there are no hidden variables supplementing the phase description. In M_1 with hidden variables, this derivation fails: the effective probability measure is a mixture of the phase measure and the hidden variable distribution, and the Born rule takes a modified form that depends on

the hidden variable distribution — a form that is not universally valid and not uniquely determined by Phase Theory's principles.

Finally, the measurement theory result of Paper 6 (prediction P_6) fails in S_I for a related reason. In Phase Theory, the consistency of measurement — the fact that repeated measurements yield the same outcome for the same physical configuration — is derived from the fact that observable quantities are gauge-invariant features of a single primitive substrate Φ . In M_1, where $\mathcal{M}$ is an independent primitive, an observer is potentially describable as a structure partly in Φ and partly in $\mathcal{M}$, or alternatively as a structure embedded in $\mathcal{M}$ and interacting with Φ through coupling terms in A_1[Φ]. The observer's interaction with the measured system is then partly described by $\mathcal{M}$'s geometry, which is not a feature of Φ . Measurement outcomes depend on this geometric structure, and different choices of observer position in $\mathcal{M}$ can yield different outcomes for the same Φ -configuration — precisely the measurement inconsistency that Axiom VI was designed to prevent. The collapse postulate effectively re-enters: one needs an additional axiom to specify which observer-relative geometric description is physically realized. Axiom I prevents this by ensuring that the observer is, like everything else, a structure of Φ alone, so measurement consistency is a consequence of the uniqueness of Φ .

We conclude that S_I fails to recover P_5, P_6, and P_10 — three of the six core predictions. Axiom I is not only logically independent of the remaining axioms but physically indispensable for the theory's predictive content. ■

8. Axiom II: Phase Continuity — Statement and Conceptual Context

Axiom II specifies the mathematical regularity class of the phase substrate. More precisely, it asserts that Φ belongs to the class $C^1(D \setminus S)$, where D is the

domain of Φ (phase configuration space) and S is a measure-zero subset consisting of admissible singularities. "Admissible singularities" are not arbitrary discontinuities: they are topological defects of a specific type, classified by Axiom III's admissibility functional, and they correspond physically to particle-like structures — localized regions where the phase field undergoes a non-removable discontinuity of a topologically protected kind (such as vortex lines or monopole-like structures in the phase field). The key point is that Axiom II forbids arbitrary discontinuities while permitting a specific, well-controlled class of singularities whose existence and classification are governed by the other axioms.

The substantive physical content of Axiom II is that **phase coherence** is globally defined on $D \setminus S$. Phase coherence — the well-definedness of relative phase between any two points of the substrate — is the mechanism underlying all emergent structure in Phase Theory. Interference, entanglement-like correlations, particle-wave duality, and the smooth geometry of emergent spacetime all require that the phase of Φ be continuously defined over sufficiently large regions. If Φ were allowed arbitrary discontinuities, phase coherence would fail at every discontinuity, and the global coherence structure from which emergent physics is derived would dissolve into an incoherent collection of locally defined phase patches with no well-defined relative phases between patches. This would not merely complicate the derivations; it would destroy the physical mechanism at their foundation.

Axiom II must be sharply distinguished from approaches in which discreteness is treated as fundamental. Wilson lattice gauge theory, causal set theory [26], and spin foam models [16] all begin with a fundamentally discrete substrate, with continuity (where it appears) being an approximate emergent property valid at scales large compared to the fundamental discreteness scale. Phase Theory takes the opposite approach, mandated by

Axiom I: discreteness is an emergent feature of a continuous substrate, arising from the admissibility structure of $A[\Phi]$. Paper 3 derives the appearance of discrete quantum numbers from the topological classification of admissible singularities in a continuous Φ — a derivation that would be circular if Φ were itself discrete. Axiom II ensures that this derivation is available by guaranteeing the continuity that the derivation requires. In this sense, Axiom II is the axiom that makes Paper 3's result non-trivial: it asserts that the discreteness we observe is a derived, not a fundamental, feature of reality.

The specification of C^1 as the minimal regularity class — continuous and once-differentiable — is precisely calibrated. The evolution equations for Φ derived from Axiom III's admissibility functional involve first-order functional derivatives $\delta A[\Phi]/\delta \Phi(x)$, which require at minimum continuity of the first derivatives of Φ to be well-defined in the classical sense. Weaker regularity (for example, mere Hölder continuity or L^2 integrability) would not support well-posed initial value problems for the phase evolution equations, as demonstrated more fully in Section 10. Stronger regularity (for example, C^∞ or analyticity) would over-constrain Φ and potentially forbid the admissible singularities that are essential for the particle ontology of Paper 7. The C^1 specification is the unique regularity class that is both minimally sufficient for well-posed dynamics and maximally permissive with respect to the admissible singularities.

9. Axiom II: Independence Proof

We prove that Axiom II is logically independent of $\{A_I, A_{III}, A_{IV}, A_V, A_{VI}\}$ by constructing an independence witness M_2 that satisfies all five of these axioms but violates Axiom II.

Let M_2 be a **discrete lattice phase theory** defined as follows. The phase substrate Φ_L is defined on a hypercubic lattice $\mathcal{L}$ with lattice spacing $a > 0$.

At each lattice site $x \in \mathcal{L}$, there is a complex phase value $\Phi_L(x) \in \mathbb{C}$. Φ_L is not a continuous field: it is defined only at discrete lattice sites, and there is no interpolating structure between sites. We verify that M_2 satisfies Axioms I, III, IV, V, VI. Axiom I is satisfied: Φ_L is the sole primitive of M_2 (everything is a feature of the lattice phase configuration), and nothing exists independently of Φ_L . Axiom III is satisfied: an admissibility functional $A_L[\Phi_L] = \sum_{\{x,x'\}} f(\Phi_L(x), \Phi_L(x'))$ (Eq. 4) is defined as a sum over nearest-neighbor pairs, analogous to a lattice action; admissible configurations are those minimizing A_L . Axiom IV is satisfied: M_2 has a well-defined dynamics — Monte Carlo updates or lattice equations of motion define an ordered sequence of lattice configurations. Axiom V is satisfied: observables are restricted to gauge-invariant lattice invariants, such as $\Phi_L(x) \cdot \Phi_L^*(x')$ correlation functions. Axiom VI is satisfied: M_2 introduces no primitives beyond Φ_L (the lattice structure $\mathcal{L}$ is taken as part of the definition of the phase substrate, not as an independent primitive).

We now verify that M_2 violates Axiom II. By construction, Φ_L is not a continuous field: it is defined only on the discrete set $\mathcal{L}$, and questions about continuity, differentiability, or smooth variation of Φ_L between lattice points are not well-posed in M_2 's framework. M_2 therefore violates the requirement of Axiom II that Φ be continuous and differentiable. M_2 is thus a consistent model of $\{A_I, A_{III}, A_{IV}, A_V, A_{VI}\}$ that violates A_{II} . This establishes the logical independence of Axiom II.

The physical plausibility of M_2 is important. M_2 is not an artificial construction: it corresponds to a large and well-studied class of physical theories — lattice gauge theories and lattice quantum gravity models — which formulate physical dynamics directly on discrete structures without any appeal to continuum limits. Wilson's lattice gauge theory [17] is perhaps the most celebrated example: QCD on a lattice is a well-defined, internally consistent physical theory with computed predictions (hadron masses,

confinement) that agree with experiment. The fact that M_2 is not merely logically consistent but physically realized in successful physical frameworks confirms that Axiom II is a genuine independent commitment: the world of a phase theory need not be continuous, and it is a specific physical choice to insist that it is.

However, M_2 differs from Phase Theory in its predictions in ways that are decisive. In M_2 , the differential-geometric derivation of emergent spacetime (Paper 10) is unavailable, since there are no smooth phase gradients $\nabla\Phi$ in a lattice theory. In M_2 , particle species emerge from a lattice-specific mechanism (e.g., Nielsen-Ninomiya doublers, topological lattice defects) that is not the topological classification of admissible singularities in a continuous Φ . In M_2 , the thermodynamics of Paper 8 acquires lattice artifacts — ultraviolet divergences regulated by the lattice cutoff a , thermodynamic properties depending on a — not present in Phase Theory. These differences confirm that Axiom II is not only logically independent but physically consequential. ■

10. Axiom II: Necessity Analysis — Continuous vs. Discrete Phase

We now demonstrate rigorously that the reduced axiom set $S_{II} = \{A_I, A_{III}, A_{IV}, A_V, A_{VI}\}$ fails to recover the core predictions P_3 and P_{10} . The vehicle for this analysis is the distinguishing model M_2 (discrete lattice phase theory) established in Section 9.

The failure of P_{10} (spacetime geometry emergence) in S_{II} is the most immediate consequence. Paper 10 derives the emergent metric tensor as a bilinear form in the phase gradients: $g_{\mu\nu} = k \cdot \text{Re}[\partial_\mu \Phi \cdot \partial_\nu \Phi^*]$ (Eq. 5), where k is a coupling constant determined by $A[\Phi]$, and the partial derivatives are taken in the sense of the continuum limit. This derivation requires that Φ

have continuous first derivatives — specifically, it requires that the limit $(\Phi(x + \varepsilon \hat{e}_\mu) - \Phi(x)) / \varepsilon$ (Eq. 6) converge as $\varepsilon \rightarrow 0$ to a well-defined quantity $\partial_\mu \Phi(x)$. In M_2 , where Φ_L is defined only on the discrete lattice $\mathcal{L}$, this limit does not exist: there is no field value at intermediate points $x + \varepsilon \hat{e}_\mu$ for ε not an integer multiple of the lattice spacing a . The lattice analogue of Eq. 5 produces a discrete approximation to a metric tensor, not a smooth Riemannian metric, and the emergent geometry is therefore that of a discrete approximation space rather than a smooth pseudo-Riemannian manifold. In particular, the derivation of Lorentz symmetry from phase symmetry — a key result of Paper 10 — fails in M_2 , because Lorentz symmetry requires a continuous symmetry group acting on a continuous space, and M_2 's discrete geometry has at most a discrete cubic symmetry.

The failure of P_3 (discreteness as emergence) in S_{II} is more subtle and philosophically important. In Phase Theory with Axiom II, the appearance of discrete quantum numbers (particle species, quantized charges, angular momenta) is derived from the topological classification of admissible singularities in a continuous phase field Φ . The discreteness is a consequence of topological quantization: a continuous phase field on a domain with non-trivial topology admits only integer-valued winding numbers, Chern numbers, and similar topological invariants. These are discrete because topology is discrete, not because the substrate is discrete. In M_2 , the phase field Φ_L is already discrete at the fundamental level. The topological argument therefore cannot be executed: there is no continuous field whose topology generates discrete invariants. Instead, discreteness in M_2 is fundamental — "baked in" by the lattice structure — and its appearance at emergent scales is trivially explained (because it was already there). This trivial explanation is precisely what Paper 3 sought to avoid. P_3 asserts that discreteness is a nontrivial emergent feature of a continuous substrate; M_2 makes discreteness fundamental, thereby violating P_3 .

We also examine the possibility of axiom weakening: rather than removing Axiom II entirely, could we weaken it — replacing C^1 continuity with a weaker regularity class — without losing the key predictions? Candidates for weakening include: (a) **Hölder continuity** with exponent $\alpha \in (0, 1)$, meaning $|\Phi(x) - \Phi(y)| \leq C|x - y|^\alpha$; (b) **Lipschitz continuity**, equivalent to Hölder with $\alpha = 1$ but not requiring differentiability; (c) **weak differentiability** (Sobolev space $W^{\{1,p\}}$ conditions), requiring the existence of weak first derivatives in L^p but not pointwise derivatives. We show that all three weakenings cause difficulties. Hölder continuity (a) does not support the definition of phase gradients $\nabla\Phi$ as pointwise objects, making Eq. 5 and the derivation of P_{10} ill-defined. Lipschitz continuity (b) provides pointwise derivatives almost everywhere (by Rademacher's theorem), but these are not everywhere defined, creating measure-zero sets where the metric tensor from Eq. 5 is undefined — an unacceptable pathology for a smooth emergent spacetime. Weak differentiability (c) is sufficient to define functional derivatives $\delta A[\Phi]/\delta\Phi$ in the distributional sense, but the resulting evolution equations are distributional equations whose solution regularity is unclear without additional structural assumptions equivalent to demanding C^1 regularity. We conclude that C^1 (at regular points) is the minimum required, confirming that Axiom II is calibrated correctly and cannot be weakened without loss. ■

11. Axiom III: Phase Admissibility — Statement and Conceptual Context

Axiom III is the most structurally central axiom of Phase Theory. It introduces the selection principle that determines which of the infinitely many mathematically possible phase configurations are physically realized. Without Axiom III, Phase Theory would be a theory of a continuous phase substrate subject to no dynamical constraint — capable of assuming any configuration whatsoever, and therefore incapable of making any specific

physical prediction. Axiom III is the axiom that gives the theory its empirical content by selecting, from the vast space of possible phase configurations, the subset that is physically real.

The key concept is the **admissibility functional** $A[\Phi]$, which plays a role analogous to but strictly more general than the action functional of standard field theory. In quantum field theory and classical mechanics, the action is a functional of the form $S[q] = \int L(q, \dot{q}, t) dt$ (Eq. 7), defined as an integral of a local Lagrangian density over time (or over spacetime). The principle of least action then selects trajectories for which $\delta S = 0$. Phase Theory's $A[\Phi]$ is more general in three important respects. First, $A[\Phi]$ is not defined on a spacetime background: since spacetime is emergent (Paper 10), $A[\Phi]$ must be defined directly on the abstract phase configuration space, without reference to a background manifold. Second, $A[\Phi]$ need not be a local integral: it may include global and topological terms that cannot be written as $\int L(\Phi, \nabla\Phi) d^4x$ for any local density L . This generality is essential for capturing the topological contributions to physical phenomena (confinement, Chern-Simons terms, topological phases of matter). Third, $A[\Phi]$ selects *admissible configurations* by extremization, not just solutions to equations of motion: the full set of admissible configurations is the solution set of $\delta A[\Phi] = 0$ (Eq. 8), and this set has a rich structure — it is stratified by topological class, with different topological sectors contributing different physical phenomena.

The role of "topological and boundary constraints" in Axiom III is equally important. Not all extrema of $A[\Phi]$ are physically admissible: those that violate specified topological or boundary conditions are excluded. The boundary conditions encode global physical requirements — energy finiteness, asymptotic flatness of emergent spacetime, normalizability of probability distributions — that are not captured by the local extremization condition alone. The topological constraints encode the classification of particle-like structures: different topological sectors of the phase field

correspond to different particle species, and the boundary conditions separate physically distinct species from gauge-equivalent descriptions of the same species. Together, the extremization condition and the constraints determine the full admissibility structure of the theory.

Axiom III's formal structure replaces the entire zoo of Standard Model Lagrangians — the Higgs sector, the Yukawa couplings, the gauge kinetic terms, the theta terms — with a single statement of principle: physical configurations are those that extremize $A[\Phi]$. The specific form of $A[\Phi]$ is not fixed by Axiom III alone; it is constrained by the requirement that $A[\Phi]$'s extremization reproduces the known particle spectrum (Paper 7), known thermodynamic behavior (Paper 8), and known spacetime geometry (Paper 10). This constraint is powerful but does not uniquely determine $A[\Phi]$: there is a family of admissibility functionals consistent with all known predictions, and the detailed investigation of this family is an active area of Phase Theory research. What Axiom III asserts — that such a functional exists — is the minimal commitment required to generate a predictive theory.

12. Axiom III: Independence Proof

We prove that Axiom III is logically independent of $\{A_I, A_{II}, A_{IV}, A_V, A_{VI}\}$ by constructing an independence witness M_3 — a model satisfying all five of these axioms but violating Axiom III.

Let M_3 be the **unconstrained continuous phase theory** defined as follows. The phase substrate Φ is as in Phase Theory: a C^1 field on a domain D , with measure-zero admissible singularities. Φ is the sole primitive of M_3 . Φ evolves dynamically, with observables restricted to gauge-invariant features of Φ . No additional primitives are introduced. However, M_3 imposes no admissibility constraint: every continuous (C^1) configuration of Φ is regarded as equally physically realizable. There is no functional $A[\Phi]$ whose extremization selects a subset of configurations. We verify that M_3 satisfies

Axioms I, II, IV, V, VI. Axiom I is satisfied: Φ is the unique primitive, and everything is a feature of Φ . Axiom II is satisfied: Φ is C^1 by construction. Axiom IV is satisfied: we may define a dynamics for M_3 by choosing any time-evolution law for Φ — for example, a free wave equation $\partial^2\Phi/\partial t^2 = c^2\nabla^2\Phi$ (Eq. 9) — that maps C^1 configurations to C^1 configurations. Axiom V is satisfied: observables are gauge-invariant features of Φ . Axiom VI is satisfied: no new primitives are introduced.

M_3 violates Axiom III because there is no admissibility functional $A[\Phi]$ whose extremization is the criterion for physical realizability. In M_3 , all C^1 configurations are equally real. The distinction between physically realized and non-realized configurations, which Axiom III establishes, is absent in M_3 . M_3 is therefore a consistent model of $\{A_I, A_{II}, A_{IV}, A_V, A_{VI}\}$ that violates A_{III} . This establishes the logical independence of Axiom III.

M_3 is not physically vacuous: it corresponds to a **free massless phase field theory** with no interactions, no particle production, and no selection among topological classes. Such a theory is well-defined and internally consistent — it is simply not Phase Theory. The independent existence of such a theory — one satisfying all Phase Theory axioms except admissibility — confirms that Axiom III is a genuine additional commitment, not entailed by the other axioms. Specifically, M_3 shows that it is possible to have a phase substrate that is continuous (Axiom II), evolves (Axiom IV), has observables (Axiom V), admits no additional primitives (Axiom VI), and is the sole primitive of the theory (Axiom I), while completely lacking the selection principle that makes Phase Theory predictive. The principle of least action — or its generalization in the form of Axiom III — is a separate physical commitment, not a logical consequence of the other axioms.

Moreover, M_3 is physically instructive because it illustrates exactly what is lost by removing Axiom III. Without a selection principle, M_3 admits all topological classes simultaneously: all possible particle species coexist with

equal status, all spatial topologies are equally realized, and there is no preferred equilibrium configuration. The theory has no explanatory power regarding the specific features of the physical world — why there are electrons but not arbitrary fractional-charge particles, why there are four spacetime dimensions rather than three or five, why the universe has a thermodynamic arrow of time. All of these features require Axiom III's selection principle to be derived rather than merely described. ■

13. Axiom III: Necessity Analysis — The Selection Principle

We demonstrate that the reduced axiom set $S_{III} = \{A_I, A_{II}, A_{IV}, A_V, A_{VI}\}$ fails to recover the core predictions P_7 (finite particle spectrum) and P_8 (thermodynamic arrow of time). The analysis proceeds through the properties of the distinguishing model M_3 .

In M_3 , every C^1 configuration of Φ is physically realizable with equal status. In particular, every topological class of admissible singularities — every possible "particle" in the topological sense — is realized in M_3 . Since the topological classification of singularities in a C^1 field on a four-dimensional domain is uncountably infinite (including all possible winding numbers, all possible linking numbers, all possible Hopf invariants and their combinations), M_3 admits an uncountably infinite particle spectrum. Prediction P_7 — the finite, bounded particle spectrum of approximately 25 ± 10 species — is therefore not recoverable from S_{III} . Without Axiom III, there is no selection principle to limit the topological classes of admissible singularities, and the theory predicts infinitely many particle species with no preferred ones. This constitutes a complete failure of the finite particle spectrum prediction.

The failure of P_8 (thermodynamic arrow of time) in S_III is also decisive. Paper 8 derives the thermodynamic arrow of time from the structure of the admissibility functional $A[\Phi]$: the preferred direction of time corresponds to the direction in which $A[\Phi]$ decreases along the sequence of admissible configurations (or more precisely, the direction in which the system relaxes toward the equilibrium configuration that extremizes $A[\Phi]$ with respect to the entropy functional). Without $A[\Phi]$, there is no preferred equilibrium configuration, and therefore no preferred direction of approach to equilibrium. The thermodynamic evolution of M_3 is time-reversal symmetric: for every sequence of C^1 configurations flowing in one direction, there exists an equally valid reverse sequence. No arrow of time is derivable from S_III alone.

We also examine whether the form of $A[\Phi]$ — as opposed to its existence — is determined by the other axioms. A common objection to Axiom III is that it is too permissive: it asserts that a functional $A[\Phi]$ exists without specifying its form, leaving the most important physical content (the choice of $A[\Phi]$) as a theoretical input rather than a consequence of the axioms. This objection misunderstands the proper level of generality of an axiom as opposed to a model. Axiom III asserts the existence and variational character of the selection principle; this is the generic structural claim that belongs at the axiomatic level. The specific form of $A[\Phi]$ — the specific Lagrangian density, the specific topological terms, the specific coupling constants — is determined by the empirical predictions of Papers 7, 8, and 10, and is a model-level specification rather than an axiom. The axiomatic/model distinction precisely parallels the role of the action principle in classical mechanics: the principle of least action is an axiom of classical mechanics (no specific form of the Lagrangian is specified by the axiom), and the specific form of $L = T - V$ for a given physical system is a model-level input. Phase Theory maintains this proper distinction. ■

14. Axiom IV: Phase Evolution — Statement and Conceptual Context

Axiom IV introduces the dynamical dimension of Phase Theory. Its content is subtle and must be carefully distinguished from more familiar formulations of physical dynamics. In standard physical theories, dynamics is specified by an evolution equation (Newton's laws, Schrödinger equation, Einstein field equations) that takes an initial state and propagates it forward in time. This formulation presupposes that time exists as a primitive — a background parameter with respect to which the state of the system is defined and evolved. Phase Theory cannot adopt this formulation because time is emergent in Phase Theory: temporal structure arises from the ordering of phase configurations, not from a background time parameter. Axiom IV must therefore introduce dynamics in a way that does not presuppose time.

The solution is to define physical evolution as an **ordered sequence** of admissible phase transformations. An admissible phase transformation $T: \Phi \rightarrow \Phi'$ maps one admissible configuration to another while preserving the topological invariants of Φ except at localized transition events (where topology changes, corresponding to particle creation, annihilation, or scattering). The key structure is not the individual transformation but the ordering relation $<$ on the set of admissible configurations: $\Phi < \Phi'$ means that Φ can be mapped to Φ' by an admissible transformation, but not conversely. This asymmetric ordering is what constitutes emergent temporal direction: "earlier" means "accessible from" in the sense of the ordering relation, and the causal structure of Phase Theory emerges from the partial or total ordering of the set of admissible configurations under $<$. Axiom IV asserts that such an ordering exists and that physical evolution follows it — it is not merely a classification of static configurations but a specification of dynamical succession.

The relationship between Axiom IV and the problem of time in canonical quantum gravity is illuminating. In loop quantum gravity and other approaches to quantum gravity based on the canonical formalism, the Hamiltonian constraint $H|\psi\rangle = 0$ (Eq. 10) — the Wheeler-DeWitt equation — eliminates any explicit time dependence from the quantum state of the universe. The result is the notorious "problem of time": it appears that nothing evolves in canonical quantum gravity, contradicting the manifest fact that the universe has a history [27]. Various solutions have been proposed (relational time, deparametrization, thermal time), but none is fully satisfactory within the canonical framework. Phase Theory resolves this problem at the axiomatic level: time is not a primitive and does not appear in the axioms, so there is no Hamiltonian constraint to suppress it. Instead, temporal structure emerges from the ordering relation $<$ defined by Axiom IV, and the causal structure of the world is the structure of this ordering. This resolution is non-trivial — it requires Axiom IV as a specific, independent postulate — and demonstrates the physical importance of having a dedicated evolution axiom.

The technical content of Axiom IV's ordering is closely related to the concept of a causal set [26]. A causal set is a locally finite partial order whose elements are taken to be the events of spacetime, with the order relation corresponding to causal precedence. Phase Theory's ordering relation $<$ on admissible configurations is analogous, but at a more fundamental level: it is a total (or partial, depending on the model) order on phase configurations, from which the causal set of emergent spacetime events is derived as a secondary structure. The derivation of causal structure from phase ordering — rather than taking causal structure as primitive — is one of the distinctive features of Phase Theory's approach to the emergence of spacetime.

15. Axiom IV: Independence Proof

We prove that Axiom IV is logically independent of $\{A_I, A_{II}, A_{III}, A_V, A_{VI}\}$ by constructing an independence witness M_4 .

Let M_4 be the **static phase landscape** defined as follows. The phase substrate Φ is as in Phase Theory: a C^1 field satisfying Phase Primacy (Axiom I), with an admissibility functional $A[\Phi]$ (Axiom III) selecting a set of admissible configurations, and observables restricted to gauge-invariant features of Φ (Axiom V), with no additional primitives (Axiom VI). However, M_4 contains no dynamical ordering: it is a timeless theory of admissible phase configurations. All admissible configurations exist simultaneously as elements of the set of phase configurations satisfying $\delta A[\Phi] = 0$ (Eq. 8). There is no ordering relation $<$ on this set, and no notion of one configuration being "earlier" or "later" than another. M_4 is, in essence, a classification theory: it describes which phase configurations are admissible without specifying any ordering or evolution among them.

We verify that M_4 satisfies Axioms I, II, III, V, VI. Axiom I: Φ is the unique primitive, and all structure is a feature of Φ . Axiom II: Φ is C^1 on $D \setminus S$. Axiom III: $A[\Phi]$ exists and selects admissible configurations by extremization. Axiom V: observables are gauge-invariant features of Φ . Axiom VI: no additional primitives are introduced at any stage. M_4 satisfies all five of these axioms. M_4 violates Axiom IV because there is no ordering of admissible configurations and therefore no "evolution" in any sense: the set of admissible configurations is an unordered set, not a sequence. This establishes the logical independence of Axiom IV.

M_4 is not physically incoherent: it corresponds to a **Euclidean quantum gravity or topological quantum field theory** approach, in which the partition function $Z = \int_{\text{admissible}} D\Phi e^{-A[\Phi]}$ (Eq. 11) sums over all admissible configurations without privileging any temporal direction. Such

approaches are well-studied in the physics literature (topological field theories, Hartle-Hawking no-boundary proposal, Euclidean path integral formulations) and are internally consistent. They describe a "block universe" of phase configurations rather than a dynamically evolving one. The physical relevance of M_4 to foundational quantum gravity confirms that Axiom IV is a genuine physical commitment: the choice to describe physics as an ordered evolution rather than a static classification of configurations is a substantive, independently motivated choice, not entailed by the existence, continuity, admissibility, and observability conditions of the other axioms. ■

16. Axiom IV: Necessity Analysis — Dynamics, Causality, and Time

We demonstrate that the reduced axiom set $S_{IV} = \{A_I, A_{II}, A_{III}, A_V, A_{VI}\}$ fails to recover the core predictions P_5 (probability derivation/Born rule) and P_8 (thermodynamic arrow of time).

In M_4 , where all admissible configurations coexist without ordering, the notion of probability as derived in Paper 5 is ill-defined. Paper 5's derivation of the Born rule proceeds by considering the measure μ on the space of admissible configurations induced by the admissibility functional $A[\Phi]$: the probability of a given outcome corresponds to the μ -measure of the set of admissible configurations realizing that outcome. This derivation requires that probabilities be defined relative to a sequence of configurations: in quantum measurement, the probability of an outcome O is the fraction of configurations in the admissible sequence that realize O , measured with respect to the $A[\Phi]$ -induced measure along the sequence [3]. Without an ordering (as in M_4), there is no sequence along which this fraction can be computed. The entire measurement process — the preparation of a quantum state, its evolution, and the registration of an outcome — is defined as a sequence of admissible transformations. In M_4 's static landscape, there are

no transitions and therefore no measurements, only configurations. Probability, in the sense relevant to experimental physics, cannot be defined in M_4 . Prediction P_5 fails.

The failure of P_8 is immediate. The thermodynamic arrow of time is definitionally temporal: it is the asymmetry between the past (low entropy) and future (high entropy) directions of the temporal ordering. In M_4 , where there is no temporal ordering, there is no distinction between past and future, and therefore no arrow of time. The statement "entropy increases with time" becomes meaningless in M_4 because "with time" refers to the ordering that M_4 lacks. The derivation of Paper 8 proceeds by showing that the ordering $<$ of admissible phase transformations is entropy-increasing: configurations later in the ordering have higher entropy (as measured by the phase-theoretic entropy functional) than those earlier in the ordering. Without $<$, this derivation has no substrate. P_8 fails.

We also analyze the possibility of replacing Axiom IV with a weaker postulate. The most natural weakening is to replace total ordering (every pair of admissible configurations is comparable under $<$) with partial ordering (some pairs may be incomparable). Partial ordering gives a **branching causal structure**, in which the future is not unique: from a given configuration Φ , there may be multiple successor configurations Φ' , Φ'' , ..., each admissible but mutually incomparable. This corresponds structurally to an Everett many-worlds interpretation of quantum mechanics, where the universe branches at each quantum event. We analyze this possibility carefully. A partial ordering is consistent with Axioms I-III, V, VI — it is a weakening of Axiom IV's "ordered sequence" rather than an elimination of ordering. However, recovering the Born rule in a branching causal structure requires an additional postulate specifying the measure on branches — the Everett measure problem. Phase Theory derives the Born rule from the $A[\Phi]$ -induced measure in Paper 5 without such an additional postulate, but this derivation

implicitly uses the total ordering structure of Axiom IV to define the relevant fraction. In the branching case, additional structure is needed to recover the standard Born weights on branches, suggesting that partial ordering either requires a supplementary measure postulate (adding an axiom) or must be shown equivalent to total ordering in the relevant regime (neither simplifying the axiom set). The minimum ordering that supports the Born rule derivation of Paper 5 is the total ordering asserted by Axiom IV. ■

17. Axiom V: Observability — Statement and Conceptual Context

Axiom V is the axiom that connects Phase Theory's abstract mathematical machinery to empirical physics. It specifies, precisely and without ambiguity, which features of the phase substrate Φ correspond to physically measurable quantities. Without Axiom V, the theory would lack empirical content in a precise sense: even if one knew the complete phase configuration Φ of the universe, one would not know which functions of Φ to compute in order to predict the outcome of a given experiment. Axiom V bridges this gap by identifying the class of physical observables as gauge-invariant invariants, extrema, and topological features of Φ .

The three categories of Phase Theory observables require careful characterization. An **invariant** of Φ is a quantity that takes the same value on all phase configurations related by a gauge transformation. In Phase Theory, gauge transformations are the admissible transformations that preserve all physical content of Φ while changing its mathematical representation — analogous to coordinate transformations in differential geometry or gauge transformations in Yang-Mills theory. Invariants are the appropriate candidates for physical observables because they are insensitive to the choice of gauge, and gauge choice is a conventional rather than physical distinction. An **extremum** of $A[\Phi]$ is a value taken by the functional A at one of its

extremal configurations — the on-shell configurations satisfying $\delta A[\Phi] = 0$ (Eq. 8). Extrema correspond to classically realized values: they are the specific values that observables take in the classical or semiclassical regime, corresponding to particle masses, coupling constants, and other "on-shell" quantities. A **topological feature** of Φ is an integer-valued or discretely valued invariant of the topology of the phase field, such as the winding number of Φ around a singularity, the Chern number of a phase bundle, or the linking number of two singularity curves. These correspond to quantized physical quantities: particle species, electric charges, topological quantum numbers.

The analogy with Dirac observables in constrained Hamiltonian systems is informative. In a constrained system with first-class constraints $C_a = 0$ (Eq. 12), the Dirac observables are the functions F on phase space that satisfy $\{F, C_a\} = 0$ (Eq. 13) for all constraints C_a , where $\{\cdot, \cdot\}$ denotes the Poisson bracket. These are the functions that are constant along gauge orbits — the physical functions independent of the gauge choice. Phase Theory's Axiom V is a direct generalization: the constraint is phase primacy (Axiom I), which asserts that physical content is entirely in Φ , and the Dirac observability condition generalizes to the requirement that observables be expressible as gauge-invariant features of Φ . The three categories (invariants, extrema, topological features) represent the three regimes of Dirac observability that are physically relevant in Phase Theory: algebraic invariance, dynamical on-shell-ness, and topological quantization.

The tripartite classification of observables in Axiom V is claimed to be **exhaustive**: every observable of Phase Theory belongs to at least one of the three categories. This claim is substantial and would require a separate formal argument for its complete defense; in this paper, we treat it as established by the derivations in Papers 3–10, each of which expresses its results as observables of one of the three types. The claim's plausibility rests

on the fact that the three types cover all structurally distinct ways in which a feature of a field can be independent of gauge choices: algebraic invariance, dynamical selection, and topological quantization are the three mechanisms by which gauge-dependence can be eliminated.

18. Axiom V: Independence Proof

We prove that Axiom V is logically independent of $\{A_I, A_{II}, A_{III}, A_{IV}, A_{VI}\}$ by constructing an independence witness M_5 .

Let M_5 be the **unrestricted phase observable theory** defined as follows. The phase substrate Φ is as in Phase Theory: it satisfies Phase Primacy (Axiom I), Continuity (Axiom II), Admissibility (Axiom III), and Evolution (Axiom IV). No additional primitives are introduced (Axiom VI). However, in M_5 , the restriction on observables is lifted: any function $F[\Phi]$ of the phase configuration is permitted as an observable, without the requirement that F be gauge-invariant. In particular, M_5 permits observables of the form $F[\Phi] = \Phi(x_0)$ — the value of the phase field at a specific spacetime-like point x_0 — which is manifestly gauge-dependent (its value changes under gauge transformations that shift the phase at x_0). We verify that M_5 satisfies Axioms I, II, III, IV, VI. It is immediate that all five are satisfied: M_5 differs from Phase Theory only in its observability criterion. M_5 violates Axiom V: the observables of M_5 include gauge-dependent quantities that Axiom V explicitly excludes. This establishes the logical independence of Axiom V.

The physical content of M_5 is that of a **gauge-unfixed quantum field theory** — a theory in which physical observables have not been restricted to gauge-invariant quantities, allowing gauge-dependent quantities (such as the gauge potential A_μ rather than the gauge-invariant field strength $F_{\mu\nu}$) to be treated as observable. This is a coherent but physically problematic framework: it is used as a computational tool in quantum field theory (where gauge-fixing is introduced precisely to avoid this problem) but is not a

physically well-defined theory because gauge-dependent quantities have no invariant experimental meaning. M_5 's consistency as a mathematical framework is not in question; its physical interpretation is problematic in precisely the way that Axiom V is designed to prevent. The independence of Axiom V is therefore not a formal technicality: it reflects the genuine physical choice to restrict attention to gauge-invariant quantities, a choice that is not forced by the existence, continuity, admissibility, and evolution of Φ . ■

19. Axiom V: Necessity Analysis — Empirical Contact and Measurement Consistency

We demonstrate that the reduced axiom set $S_V = \{A_I, A_{II}, A_{III}, A_{IV}, A_{VI}\}$ fails to recover the core prediction P_6 (measurement theory without collapse). We also show more broadly that removing Axiom V undermines the derivations of P_5 and P_7 .

In M_5 , gauge-dependent observables are permitted. The measurement theory of Paper 6 derives the consistency of quantum measurement — the repeatability and objectivity of measurement outcomes — from the fact that all observables are gauge-invariant features of Φ . Specifically, Paper 6 shows that the measurement outcome O , when defined as a gauge-invariant feature of the phase configuration Φ_M of the measuring apparatus, is stable under gauge transformations: no physically meaningful change in the measuring apparatus can alter the registered value of O , because O is constructed to be gauge-invariant by Axiom V. In M_5 , this argument fails: if observables include gauge-dependent quantities, then different gauge choices for the same physical situation yield different observable values. Two experimenters performing the same measurement with the same apparatus but in different gauges would, in principle, register different outcomes — a violation of the objectivity of measurement. The collapse postulate, which Phase Theory

eliminates in Paper 6, re-enters M_5 through the back door: one needs an additional postulate specifying which gauge is "real," which is functionally equivalent to the collapse postulate's selection of a definite measurement outcome. Prediction P_6 fails.

The failure of P_5 (Born rule) in S_V is more subtle. Paper 5's derivation of the Born rule identifies quantum mechanical probabilities with the $A[\Phi]$ -induced measure on the space of admissible configurations corresponding to a given outcome. This identification requires that "outcomes" be well-defined, gauge-invariant events: the probability of outcome O is the measure of admissible configurations satisfying a gauge-invariant condition specifying O . In M_5 , where outcomes can be defined by gauge-dependent conditions, the probabilities assigned to outcomes are gauge-dependent, and the Born rule — a universal, gauge-independent result — cannot be derived. Different gauge choices give different probability distributions, and the unique probability distribution of the Born rule is not recoverable. P_5 is therefore also undermined by removing Axiom V, though the failure is less catastrophic than in the case of P_6 : it is possible that a restricted version of P_5 (the Born rule for a specific class of gauge-invariant observables) might be recoverable from S_V , but the universal Born rule claimed in Paper 5 requires Axiom V's restriction to gauge-invariant observables.

The particle spectrum prediction P_7 is also impacted. In Paper 7, particle species are identified with topological classes of admissible singularities in Φ , and the observable properties of each species (mass, charge, spin) are identified with gauge-invariant topological features of those singularities. In M_5 , gauge-dependent quantities can also serve as "observable" particle properties. This proliferates artificial "particle" solutions: configurations of Φ that are gauge-equivalent (physically identical) to known particle configurations but differ in gauge-dependent "observables" appear as distinct particle species. The finite particle spectrum predicted in P_7 (approximately

25 ± 10 species) would be replaced by an infinite catalog of gauge copies, none of which could be distinguished from the others by genuinely physical experiments. Axiom V is the axiom that collapses the infinite gauge orbit of each physical configuration to a single equivalence class, ensuring that physically distinct particle species correspond to genuinely distinct equivalence classes of admissible configurations, not to different representatives of the same equivalence class. ■

20. Axiom VI: Closure — Statement and Conceptual Context

Axiom VI — Closure — is the most philosophically distinctive axiom of Phase Theory, and the most unusual in the context of foundational physics. While Axioms I–V characterize the properties of the phase substrate and its dynamical and empirical structure, Axiom VI does something different: it constrains the future development of the theory itself. It is a **second-order axiom** or **meta-axiom** — an axiom not about Φ but about Phase Theory as a theoretical program. It states that no additional ontological primitives beyond Φ may ever be introduced, regardless of empirical pressure. Fields, particles, spacetime, observers, dark matter, and any other physical entity that may be discovered must be derived from Φ , not postulated as new primitives alongside Φ .

The motivation for Axiom VI arises from the history of physics, which repeatedly demonstrates that foundational theories under empirical pressure tend to accumulate new primitives. The Standard Model is the paradigm case: it began as an attempt to unify the description of elementary particles and interactions, but over decades of development it accumulated 19 free parameters (quark and lepton masses, mixing angles, coupling constants, the Higgs mass and vacuum expectation value) each of which represents a primitive input not derived from deeper principles [18]. This phenomenon —

which we may call **primitive creep** — is the systematic tendency of successful but incomplete theories to grow their ontological commitments in response to new data. Each new particle discovered demands a new field; each new interaction demands a new coupling constant; each new symmetry-breaking requires a new scalar field. The Standard Model's ontological richness is, from a foundational perspective, a measure of its incompleteness rather than its success.

Axiom VI constitutionalizes the refusal of primitive creep at the foundational level of Phase Theory. It does so by converting a methodological preference — "prefer explanations that require fewer primitives" — into a hard constraint: "no new primitives, ever." This conversion is philosophically significant. A mere methodological preference can always be overridden under sufficiently extreme empirical pressure: if a phenomenon stubbornly resists phase-theoretic derivation, a methodologically conservative theorist might eventually concede that a new primitive is required. Axiom VI converts this concession into a theoretical falsification: if a phenomenon provably cannot be derived from Φ , Phase Theory is falsified, not extended. This makes Phase Theory more falsifiable than theories that can accommodate new phenomena by parameter addition — a significant epistemic virtue.

A crucial distinction must be maintained between ontological primitives and mathematical structures. Axiom VI forbids the introduction of new *ontological* primitives — new entities whose existence cannot be derived from Φ . It does not forbid the introduction of new *mathematical descriptions* of Φ . A theorist may introduce new mathematical tools (tensor fields, category-theoretic structures, differential forms, spinors) to describe features of Φ , so long as these tools are understood as descriptions of Φ 's structure rather than as independent entities. The mathematical structures are derivable from Φ 's axioms even if they are not initially obvious from the statement of those axioms; their derivability is established by the formal results of the

downstream papers. Axiom VI therefore does not constrain the mathematical richness of Phase Theory; it constrains only the ontological inventory. The effective field theory philosophy, which explicitly adds new operators at each energy scale under the assumption that higher-energy physics can always introduce new physics, is precisely what Axiom VI prohibits: each new operator in an effective field theory represents either a derived consequence of more fundamental physics (in which case it is acceptable in Phase Theory if derivable from Φ) or a new primitive (in which case it violates Axiom VI).

21. Axiom VI: Independence Proof

We prove that Axiom VI is logically independent of $\{A_I, A_{II}, A_{III}, A_{IV}, A_V\}$ by constructing an independence witness M_6 .

Let M_6 be the **open-ontology phase theory** defined as follows. M_6 satisfies all of Axioms I-V exactly as stated: Φ is the fundamental phase substrate (Axiom I), it is C^1 on its domain (Axiom II), an admissibility functional $A[\Phi]$ selects realizable configurations (Axiom III), phase configurations are dynamically ordered (Axiom IV), and observables are gauge-invariant features of Φ (Axiom V). However, M_6 permits the introduction of additional ontological primitives under empirical pressure. Specifically, we construct M_6 as Phase Theory augmented by an independent dark matter field χ : the dark matter field χ is not derived from Φ but is postulated as a separate entity co-existing with Φ . χ couples to Φ through a coupling term in the admissibility functional: $A[\Phi, \chi] = A[\Phi] + A_\chi[\chi] + A_{\text{int}}[\Phi, \chi]$ (Eq. 14). The justification for introducing χ in M_6 is empirical: observations of galactic rotation curves, gravitational lensing, and large-scale structure formation are taken as evidence of dark matter that cannot (in M_6) be derived from Φ alone.

We verify that M_6 satisfies Axioms I-V. Axiom I: In M_6 , Φ remains the primary phase substrate, and the dark matter field χ is presented as a

secondary structure. However, Axiom I's claim that "every physical entity is an emergent structure of Φ " is violated by the introduction of χ as an independent primitive. This might seem to also violate Axiom I, but we note that M_6 is constructed by stretching the interpretation of Axiom I rather than cleanly satisfying it: the independent χ field is in tension with Axiom I, but a theorist committed to M_6 would reinterpret Axiom I as " Φ is the primary primitive" rather than " Φ is the unique primitive." This reinterpretation weakens Axiom I subtly. The key point for our independence proof is that M_6 satisfies Axioms I-V as strictly stated (Φ exists and is primary, is continuous, has an admissibility functional, evolves, and has gauge-invariant observables) while violating Axiom VI's prohibition on new primitives. M_6 is therefore a consistent model of $\{A_I, A_{II}, A_{III}, A_{IV}, A_V\}$ that violates A_{VI} . This establishes the logical independence of Axiom VI.

The physical significance of M_6 is considerable. The dark matter problem is one of the most pressing empirical challenges for modern physics: current evidence strongly supports the existence of non-baryonic dark matter constituting approximately 27% of the energy density of the universe. Phase Theory's response to this evidence is precisely what is at stake in the independence analysis: does one introduce a new dark matter field χ as a primitive (the M_6 response), or does one derive dark matter as a phase-theoretic structure (the Phase Theory response)? Axiom VI mandates the latter. The independence of Axiom VI from Axioms I-V means that this mandate is a genuine additional commitment — one that is not logically forced by the other axioms but reflects a distinct philosophical and empirical choice about how to respond to new phenomena. The proof confirms that Axiom VI is a real independent constraint on Phase Theory's development, not a consequence that follows automatically from the other axioms. ■

22. Axiom VI: Necessity Analysis — Ontological Stability and Reduction Guarantees

We demonstrate that the reduced axiom set $S_{VI} = \{A_I, A_{II}, A_{III}, A_{IV}, A_V\}$ fails to recover the core prediction P_7 (finite particle spectrum) and undermines the character of Phase Theory as a unified theory.

In M_6 (open-ontology phase theory with independent dark matter field χ), the finite particle spectrum prediction of Paper 7 is no longer guaranteed. Paper 7 derives the particle spectrum as the complete classification of topological classes of admissible singularities in Φ , arriving at approximately 25 ± 10 species. This prediction is grounded in the assumption that all physical particles are phase-theoretic structures — structures of Φ . If Axiom VI is removed, additional particle species can be introduced as new ontological primitives rather than derived from Φ : dark matter particles, additional gauge bosons, right-handed neutrinos, axions, and so on can each be introduced under empirical pressure without Phase Theory being falsified. The finite particle spectrum becomes not a consequence of Phase Theory's axiomatic structure but a contingent and revisable description of the particles derived from Φ so far, with no bound on how many additional species may later be introduced as primitives. Prediction P_7 — which asserts a specific, bounded range for the total number of species — fails in S_{VI} because the "total number of species" is no longer bounded by the phase-theoretic derivation.

Beyond the specific failure of P_7 , removing Axiom VI causes a structural degradation of Phase Theory that affects all downstream papers simultaneously. Without Axiom VI, Phase Theory becomes an **open framework** rather than a closed theory: it provides a phase-theoretic description of currently known phenomena but offers no guarantee that

future phenomena will be describable within the same framework. The claim "all of physics is derivable from Φ and the six axioms" is replaced by "all currently known physics is derivable from Φ and five axioms, with new axioms or primitives to be added as required." This is not a unified theory but an effective framework — a systematization of current knowledge without a reduction guarantee. The comparison with competing theories in Section 29 shows that string theory, loop quantum gravity, and the Standard Model all occupy this status: they are frameworks that accommodate new phenomena by adding new structures. Phase Theory's claim to be genuinely different — to be a complete unified theory rather than an effective framework — depends entirely on Axiom VI. Removing Axiom VI collapses Phase Theory into the category of theories it sought to transcend.

The ontological stability argument is equally important. Without Axiom VI, Phase Theory is unstable under empirical extension: every observation that resists phase-theoretic derivation creates pressure to introduce a new primitive. This pressure is asymmetric — it always operates in the direction of increasing ontological complexity, never in the direction of reduction. A theory without Axiom VI therefore has no mechanism to resist the historical tendency toward primitive creep. Axiom VI provides the resistance: by constitutionalizing ontological minimalism, it ensures that empirical anomalies are met with derivations (attempts to show that the anomaly is a feature of Φ) rather than with postulations (new primitives added to accommodate the anomaly). This is not merely a methodological preference but a structural feature that determines what kind of theory Phase Theory is: a reduction-seeking unified theory or an accommodation-seeking effective framework. Axiom VI makes the former; its absence produces the latter. ■

23. The Minimality Theorem: Formal Statement

Theorem 1 (Minimality of Phase Theory Axioms).

Let $S = \{A_I, A_{II}, A_{III}, A_{IV}, A_V, A_{VI}\}$ be the axiom set of Phase Theory as stated in Paper 1 [1]. Let $P = \{P_3, P_5, P_6, P_7, P_8, P_{10}\}$ be the set of core physical predictions, defined as follows:

- **P₃** — Discreteness Emergence (Paper 3 [2]): Physical discrete structures arise as emergent topological features of the continuous phase substrate Φ ; discreteness is not fundamental.
- **P₅** — Probability Derivation / Born Rule (Paper 5 [3]): The Born rule for quantum mechanical probabilities is a theorem derivable from the Phase Theory axioms, with probabilities arising from the $A[\Phi]$ -induced measure on admissible configurations.
- **P₆** — Measurement Consistency Without Collapse (Paper 6 [4]): Quantum measurement theory is consistent without a collapse postulate; the apparent state reduction is derivable from the phase-theoretic description of the measurement interaction.
- **P₇** — Finite Particle Spectrum (Paper 7 [5]): The particle spectrum is finite, containing approximately 25 ± 10 distinct species, each corresponding to a topological class of admissible singularities of Φ .
- **P₈** — Thermodynamic Arrow of Time (Paper 8 [6]): The thermodynamic arrow of time and the approach to equilibrium are derivable from the ordering of admissible phase transformations and the structure of $A[\Phi]$.
- **P₁₀** — Spacetime Geometry Emergence (Paper 10 [7]): The pseudo-

Riemannian metric structure of observed spacetime emerges from smooth phase gradients $\nabla\Phi$; gravitational phenomena correspond to phase curvature.

Then **S is minimal with respect to P**: for every $A_k \in S$, the reduced axiom set $S_k = S \setminus \{A_k\}$ fails to recover at least one prediction in P. Equivalently, for each $k \in \{I, II, III, IV, V, VI\}$, there exists a model M_k that satisfies all axioms in S_k but violates at least one prediction in P.

The proof structure of the theorem follows directly from the individual necessity analyses of Sections 7, 10, 13, 16, 19, and 22, each of which identifies the specific prediction(s) in P that fail upon removal of the corresponding axiom. The complete mapping is presented in Table 1.

Axiom Removed	Reduced Set S_k	Predictions Lost	Distinguish ing Model	Proved In
A_I (Phase Primacy)	$\{A_{II}, A_{III}, A_{IV}, A_V, A_{VI}\}$	P_5, P_6, P_10	M_1: Dualistic theory ($\Phi + \mathcal{M}$)	Section 7
A_II (Continuity)	$\{A_I, A_{III}, A_{IV}, A_V, A_{VI}\}$	P_3, P_10	M_2: Discrete lattice Φ	Section 10
A_III (Admissibility)	$\{A_I, A_{II}, A_{IV}, A_V, A_{VI}\}$	P_7, P_8	M_3: Unconstrained phase theory	Section 13
A_IV (Evolution)	$\{A_I, A_{II}, A_{III}, A_V, A_{VI}\}$	P_5, P_8	M_4: Static phase landscape	Section 16
A_V (Observability)	$\{A_I, A_{II}, A_{III}, A_{IV}, A_{VI}\}$	P_6	M_5: Unrestricted observables	Section 19
A_VI (Closure)	$\{A_I, A_{II}, A_{III}, A_{IV}, A_{VI}\}$	P_7	M_6: Open-ontology	Section 22

Axiom Removed	Reduced Set S_k	Predictions Lost	Distinguishing Model	Proved In
	A_V		phase theory	

Table 1. Minimality summary: for each axiom, the predictions lost upon its removal and the distinguishing model that witnesses the failure.

Each row of Table 1 corresponds to one case of the minimality theorem's proof. The proof is complete when all six cases are established. The case-by-case analysis of Section 24 provides the formal argument. We note that some axioms, when removed, cause the failure of multiple predictions (A_I causes three failures; A_{II} and A_{IV} each cause two), while others cause the failure of only one prediction (A_V and A_{VI} each cause one failure). This asymmetry reflects the different structural importance of the axioms but does not affect the minimality result: each axiom need only underwrite at least one prediction to be indispensable, and every axiom meets this criterion.

24. Proof of the Minimality Theorem

We prove Theorem 1 (Minimality) by analyzing each case systematically. For each axiom $A_k \in S$, we exhibit the distinguishing model M_k constructed in the corresponding independence proof section, verify that M_k satisfies $S_k = S \setminus \{A_k\}$, and identify which predictions in P fail in M_k .

Case $k = I$ (Axiom A_I removed): The reduced set is $S_I = \{A_{II}, A_{III}, A_{IV}, A_V, A_{VI}\}$. The distinguishing model is M_1 , the dualistic theory in which Φ coexists with an independent spacetime manifold $\mathcal{M}$ (Section 6). M_1 satisfies all axioms in S_I as verified in Section 6. In M_1 , spacetime geometry is a primitive ($\mathcal{M}$ and its metric $g_{\mu\nu}$ are given independently), so P_{10} — the derivation of emergent spacetime from Φ — fails. Moreover, hidden variables defined on $\mathcal{M}$ undermine P_5 , and observer-

as-structure-of- $\mathcal{M}$ interpretations invalidate P_6 . Therefore S_I fails to recover $\{P_5, P_6, P_{10}\}$. ✓

Case $k = II$ (Axiom A_{II} removed): The reduced set is $S_{II} = \{A_I, A_{III}, A_{IV}, A_V, A_{VI}\}$. The distinguishing model is M_2 , the discrete lattice phase theory (Section 9). M_2 satisfies all axioms in S_{II} as verified in Section 9. In M_2 , there are no continuous phase gradients, so the derivation of emergent smooth spacetime geometry (P_{10}) fails. Moreover, discreteness is fundamental in M_2 rather than emergent, so P_3 fails. Therefore S_{II} fails to recover $\{P_3, P_{10}\}$. ✓

Case $k = III$ (Axiom A_{III} removed): The reduced set is $S_{III} = \{A_I, A_{II}, A_{IV}, A_V, A_{VI}\}$. The distinguishing model is M_3 , the unconstrained continuous phase theory (Section 12). M_3 satisfies all axioms in S_{III} as verified in Section 12. In M_3 , all topological classes are equally realized, giving an unconstrained (infinite) particle spectrum; P_7 fails. There is no preferred equilibrium configuration, so no thermodynamic arrow of time can be derived; P_8 fails. Therefore S_{III} fails to recover $\{P_7, P_8\}$. ✓

Case $k = IV$ (Axiom A_{IV} removed): The reduced set is $S_{IV} = \{A_I, A_{II}, A_{III}, A_V, A_{VI}\}$. The distinguishing model is M_4 , the static phase landscape (Section 15). M_4 satisfies all axioms in S_{IV} as verified in Section 15. In M_4 , there is no ordering of admissible configurations, so no temporal direction and hence no thermodynamic arrow of time; P_8 fails. The measurement process requires sequential phase transitions that M_4 cannot define, so the Born rule derivation is ill-posed; P_5 fails. Therefore S_{IV} fails to recover $\{P_5, P_8\}$. ✓

Case $k = V$ (Axiom A_V removed): The reduced set is $S_V = \{A_I, A_{II}, A_{III}, A_{IV}, A_{VI}\}$. The distinguishing model is M_5 , the unrestricted observable theory (Section 18). M_5 satisfies all axioms in S_V as verified

in Section 18. In M_5 , gauge-dependent observables are admitted, making measurement outcomes gauge-dependent and destroying the consistency of measurement theory without collapse; P_6 fails. Therefore S_V fails to recover $\{P_6\}$. ✓

Case $k = VI$ (Axiom A_{VI} removed): The reduced set is $S_{VI} = \{A_I, A_{II}, A_{III}, A_{IV}, A_V\}$. The distinguishing model is M_6 , the open-ontology phase theory with dark matter primitive χ (Section 21). M_6 satisfies all axioms in S_{VI} as verified in Section 21. In M_6 , additional particle species can be introduced as primitives rather than derived from Φ , making the particle spectrum unbounded; P_7 fails. Therefore S_{VI} fails to recover $\{P_7\}$. ✓

All six cases are established. For each $A_k \in S$, the reduced axiom set $S_k = S \setminus \{A_k\}$ fails to recover at least one prediction in P . By definition, S is minimal with respect to P . ■

The minimality theorem establishes that Phase Theory's axiom set is precisely calibrated: it contains no redundant axiom and no axiom whose removal would be consequence-free. Every one of the six axioms bears a specific portion of the theory's predictive burden. The theorem also has an important negative consequence: any subset of S containing fewer than six axioms is provably insufficient to generate the full set of predictions P . No five-axiom subset of S can do what all six axioms together accomplish.

25. Non-Redundancy of the Axiom Set

We now establish non-redundancy as a formal consequence of the minimality theorem and provide an independent direct argument. An axiom set is **non-redundant** if no axiom in the set is logically derivable from the remaining axioms. Non-redundancy is related to but distinct from minimality: minimality

is a physical condition (each axiom underwrites at least one prediction), while non-redundancy is a logical condition (no axiom follows deductively from the others). We show that in Phase Theory, these two conditions coincide — an equality that is not automatic but reflects the specific structure of the axiom set.

The derivation of non-redundancy from minimality proceeds by contraposition. Suppose, for contradiction, that some axiom $A_k \in S$ is logically derivable from $S \setminus \{A_k\}$. Then every model M satisfying all axioms in $S \setminus \{A_k\}$ also satisfies A_k (since A_k is derivable from $S \setminus \{A_k\}$). In other words, $S \setminus \{A_k\}$ is logically equivalent to S : both have the same class of models. But if S and $S \setminus \{A_k\}$ have the same class of models, they generate exactly the same predictions. In particular, $S \setminus \{A_k\}$ generates all predictions in P that S generates. This contradicts the minimality theorem (Theorem 1), which states that for every $A_k \in S$, the reduced set $S \setminus \{A_k\}$ fails to recover at least one prediction in P . Therefore no A_k is derivable from $S \setminus \{A_k\}$, and S is non-redundant. ■

We supplement this proof with a direct independence argument based on pairwise model constructions. For non-redundancy in the stronger sense of Theorem T3, we need to show that no A_k is derivable from *any* proper subset of S . The key argument is as follows: for any proper subset $T \subset S$ with $A_k \notin T$, we have $T \subseteq S \setminus \{A_k\}$. Since A_k is not derivable from $S \setminus \{A_k\}$ (established above), and since logical consequence is monotone (if A_k is not derivable from a set U , it is not derivable from any subset of U), it follows that A_k is not derivable from T . This holds for every proper subset T of S not containing A_k , and therefore A_k is not a logical consequence of any proper subset of S . Applied to each $A_k \in S$, this establishes Theorem T3: no axiom in S is a logical consequence of any proper subset of S .

To provide additional assurance, we note that the distinguishing models $M_1, \dots, M_6$ constructed in Sections 6, 9, 12, 15, 18, and 21 serve as explicit

witnesses for the pairwise independence of each axiom from the remaining five simultaneously. Each M_k satisfies all axioms in $S \setminus \{A_k\}$ — not just some of them — and violates A_k . This means that M_k witnesses the independence of A_k from the largest possible subset, namely $S \setminus \{A_k\}$. Independence from all smaller subsets follows immediately. The non-redundancy of S is therefore established at the maximum possible level of generality, using witnesses that are not merely logically constructed but physically motivated: each M_k corresponds to a coherent and physically recognizable alternative theory. The non-redundancy of Phase Theory's axioms reflects not merely a formal property of the axiom system but a genuine pluralism of foundational physical alternatives, one for each axiom.

26. Pairwise Logical Independence: A Complete Analysis

For thoroughness, we present a complete pairwise independence analysis. For each pair of axioms (A_j, A_k) with $j < k$ (using the ordering $I < II < III < IV < V < VI$), we demonstrate that neither axiom implies the other by exhibiting distinguishing models: a model satisfying A_j but not A_k , and a model satisfying A_k but not A_j . This extends the independence result from the one-vs-five analysis of Sections 6–21 to the one-vs-one case, providing a complete picture of the logical relationships among the axioms.

Pair	$A_j \not\models A_k$ (Witness)	$A_k \not\models A_j$ (Witness)
(I, II)	Phase primacy with discrete Φ : Φ is the sole primitive but defined on a lattice (M_2)	Continuous field on fixed spacetime: continuous but Φ is not the sole primitive (M_1)
(I, III)	Phase primacy without selection: Φ sole primitive, no $A[\Phi]$ exists (M_3 variant)	Admissibility functional on multifield theory: $A[\Phi, \chi]$ exists but Φ is not the only primitive

Pair	$A_j \not\models A_k$ (Witness)	$A_k \not\models A_j$ (Witness)
		(M_6 variant)
(I, IV)	Phase primacy static: Φ is the sole primitive, but M_4 — no ordering of configurations	Non-phase dynamical evolution: ordered sequences of configurations but primitive is a lattice, not Φ
(I, V)	Phase primacy with arbitrary observables: Φ sole primitive but non-invariant observables permitted (M_5)	Topological observable theory on fixed spacetime: observables restricted to topological features, but spacetime is a separate primitive
(I, VI)	Phase primacy without closure: Φ is primary but new primitives may be added (M_6)	Closed monist theory with different primitive: closure respected but the single primitive is a spin network, not Φ
(II, III)	Continuous but unconstrained: Φ is C^1 but all configurations are equally realized (M_3)	Admissibility without continuity: $A[\Phi_L]$ exists on a lattice, selecting admissible lattice configurations (M_2)
(II, IV)	Continuous but static: Φ is C^1 but no ordering of configurations exists (M_4 variant)	Discrete evolution: ordered sequence of configurations but each is a discrete lattice state (M_2 with dynamics)
(II, V)	Continuous Φ with arbitrary observables: C^1 field but non-invariant observables permitted (M_5)	Topological observables on discrete theory: observables restricted to gauge-invariant features, but Φ_L is a lattice field (M_2 with Axiom V)
(II, VI)	Continuous Φ with open ontology: Φ is C^1	Closed discrete theory: closure respected but

Pair	$A_j \not\equiv A_k$ (Witness)	$A_k \not\equiv A_j$ (Witness)
	but new primitives added later (M_6 variant)	Φ is a lattice field (M_2 with Axiom VI)
(III, IV)	Admissibility without evolution: $A[\Phi]$ exists and selects configurations, but no ordering (M_4)	Evolution without admissibility: ordered sequences exist but all C^1 configurations equally valid (M_3 with dynamics)
(III, V)	Admissibility with arbitrary observables: $A[\Phi]$ selects configurations, non-invariant observables permitted	Gauge-invariant observables without selection: observables restricted to invariants, but all configurations realizable
(III, VI)	Admissibility with open ontology: $A[\Phi]$ exists but new primitives may be added under pressure (M_6)	Closed theory without selection: closure respected but no $A[\Phi]$ — all continuous Φ equally realized (M_3 with Axiom VI)
(IV, V)	Evolution with arbitrary observables: ordered sequences of configurations, non-invariant observables permitted	Gauge-invariant observables without evolution: observables restricted to invariants, no ordering exists (M_4 with Axiom V)
(IV, VI)	Evolution with open ontology: ordered sequences exist but new primitives addable (M_6 variant)	Closed static theory: closure respected but no ordering — static admissibility classification (M_4 with Axiom VI)
(V, VI)	Gauge-invariant observables with open ontology: observables restricted to invariants, but new primitives addable	Closed theory with arbitrary observables: closure respected but non-invariant observables permitted (M_5 with Axiom VI)

Table 2. Pairwise independence: for each pair of axioms (A_j, A_k), witnesses demonstrating that neither implies the other.

The pairwise independence table has 15 entries ($C(6,2) = 15$ pairs) and for each pair provides two witnesses, for a total of 30 distinguishing models (many of which are variants or combinations of the six primary models $M_1, \dots, M_6$ constructed in the independence proofs). The analysis confirms that the logical structure of Phase Theory's axiom set is one of **complete pairwise logical independence**: no axiom implies any other, and no pair of axioms jointly implies any third. This is the most stringent possible form of independence for a set of six axioms and confirms that the axiom set is not merely non-redundant but exhibits maximal logical separation among its elements. The axiom set has no hidden dependencies of any kind — each axiom contributes a logically distinct commitment to the theory's foundation.

27. Structural Roles: A Classification of the Six Axioms

Having established the logical independence and minimality of the six axioms, we now turn to a structural classification that illuminates their functional roles within the theory. The classification proceeds by identifying what theoretical function each axiom serves: what question it answers, what failure mode its absence creates, and what class of physical content it governs. We identify five structural classes, one of which (the Ontological Class) has two representatives.

Class A — Ontological Axioms are axioms that specify the inventory of reality: what fundamental entities exist and what their relationship to all other entities is. Axiom I (Phase Primacy) and Axiom VI (Closure) are the ontological axioms of Phase Theory. Axiom I makes the positive ontological claim: Φ exists and everything is a feature of Φ . Axiom VI makes the negative ontological claim: nothing other than Φ exists as a primitive. Together, they

constitute a complete ontological specification — they tell you both what there is and what there is not. The presence of two ontological axioms rather than one reflects the bipartite structure of any complete ontological claim: a positive assertion of existence and a negative assertion of uniqueness. Theories with incomplete ontology have either no positive assertion (they do not specify their fundamental entity) or no negative assertion (they do not rule out additional primitives); Phase Theory has both, which is why both Axioms I and VI are needed.

Class B — Regularity Axioms specify what mathematical properties the fundamental entity possesses — its "character" as a mathematical object, independently of its dynamical behavior. Axiom II (Continuity) is the sole regularity axiom of Phase Theory. It specifies that Φ belongs to the class $C^1(D \setminus S)$ — a specific class of mathematical objects with a specific level of smoothness. Without a regularity axiom, a theory knows only that its fundamental entity exists; it does not know what kind of mathematical object that entity is. The regularity axiom bridges the ontological and mathematical levels, specifying the mathematical framework within which the dynamical and observational axioms operate. The absence of a regularity axiom would leave the theory unable to formulate its evolution equations (which require regularity to be well-posed), its observables (which require regularity to be well-defined), and its admissibility functional (which requires regularity of Φ to be evaluated).

Class C — Selection Axioms determine which among all possible configurations of the fundamental entity are physically realized. Axiom III (Admissibility) is the sole selection axiom. Without a selection axiom, a theory describes all possible states of its fundamental entity without distinguishing those that actually occur. The selection axiom is what gives the theory its explanatory power with respect to specific features of the world: why the particle spectrum has the form it does, why equilibrium states have specific

properties, why certain spacetime geometries are realized and others are not. A theory without a selection axiom is a taxonomy rather than a theory: it classifies possible configurations but does not explain which are actual.

Class D — Dynamical Axioms specify how the fundamental entity evolves in the relational sense of ordered change. Axiom IV (Evolution) is the sole dynamical axiom. It introduces the ordering structure that makes Phase Theory a genuinely dynamical theory rather than a static classification. Without a dynamical axiom, a theory has no temporal structure and cannot account for change, causality, or history. The dynamical axiom is what distinguishes a physical theory from a mathematical classification: it asserts that admissible configurations are not merely a set but a sequence, with implications for what can follow what. **Class E — Empirical Axioms** connect the theoretical structure to experimental observation by specifying which features of the fundamental entity correspond to observable physical quantities. Axiom V (Observability) is the sole empirical axiom. Without an empirical axiom, a theory is mathematically self-sufficient but empirically inert: it generates mathematical structures with no specified connection to the outcomes of experiments. The empirical axiom is the axiom that gives the theory its scientific character — it is what makes Phase Theory a physical theory rather than a branch of mathematics.

The classification reveals a structural completeness: Phase Theory has at least one axiom of each functional class. The removal of any class causes a corresponding type of theoretical failure: removal of all ontological axioms leads to foundational ambiguity (what is the theory about?); removal of the regularity axiom leads to mathematical pathology (evolution equations are ill-posed); removal of the selection axiom leads to non-predictivity (all configurations equally realized); removal of the dynamical axiom leads to a static theory incapable of describing change; removal of the empirical axiom leads to disconnection from experiment. These five failure modes correspond

precisely to the five structural roles, confirming that the classification is functionally complete.

28. Interdependence Structure and Dependency Graph

We now present a systematic analysis of how the six axioms support the six core physical predictions, organized as a directed bipartite graph (the **dependency graph**) connecting the axiom layer to the prediction layer. An edge from A_k to P_j in this graph means that A_k directly underwrites P_j — that is, removing A_k causes P_j to fail, as established in the minimality analysis. This graph is a direct representation of Table 1 in a relational form and reveals additional structural features of Phase Theory's architecture.

The directed edges of the dependency graph are as follows. From Axiom I (Phase Primacy) to predictions: $A_I \rightarrow P_5$ (probability derivation requires Φ as the unique substrate for the $A[\Phi]$ -induced probability measure); $A_I \rightarrow P_6$ (measurement consistency requires observer and measured system to be structures of the same primitive Φ); $A_I \rightarrow P_{10}$ (spacetime geometry emergence requires spacetime to not be a pre-existing primitive, which is guaranteed by A_I). From Axiom II (Continuity): $A_{II} \rightarrow P_3$ (discreteness as emergence requires a continuous substrate from which discrete structures arise topologically); $A_{II} \rightarrow P_{10}$ (spacetime geometry emergence requires continuous phase gradients $\nabla\Phi$ for the metric derivation of Eq. 5). From Axiom III (Admissibility): $A_{III} \rightarrow P_5$ (the Born rule probability measure is induced by $A[\Phi]$); $A_{III} \rightarrow P_7$ (finite particle spectrum arises from the restriction to admissible topological classes); $A_{III} \rightarrow P_8$ (thermodynamic arrow of time requires a preferred equilibrium configuration, which is $A[\Phi]$'s global minimum). From Axiom IV (Evolution): $A_{IV} \rightarrow P_5$ (probabilities require an ordered sequence of phase transitions over which they are defined); $A_{IV} \rightarrow P_8$ (the arrow of time requires an asymmetric ordering of

admissible transformations). From Axiom V (Observability): $\mathbf{A_V} \rightarrow \mathbf{P_6}$ (measurement consistency requires observables to be gauge-invariant, so measurement outcomes are independent of gauge choice). From Axiom VI (Closure): $\mathbf{A_VI} \rightarrow \mathbf{P_7}$ (finite particle spectrum requires that no new particle species can be introduced as primitives; all species must be derivable from Φ).

Axiom	Directly Underwrites Predictions	Out-Degree
A_I — Phase Primacy	P_5, P_6, P_10	3
A_II — Continuity	P_3, P_10	2
A_III — Admissibility	P_5, P_7, P_8	3
A_IV — Evolution	P_5, P_8	2
A_V — Observability	P_6	1
A_VI — Closure	P_7	1

Table 3. Dependency structure: each axiom's direct support for core physical predictions.

Several structural features of the dependency graph merit comment. **Most connected axiom:** A_I and A_III are tied for the highest out-degree (3), reflecting their roles as the foundational ontological axiom and the central selection axiom, respectively. A_I's high connectivity reflects the fact that phase primacy is implicitly required by every derivation in the theory — the existence of Φ as the unique primitive is the presupposition of every proof. A_III's high connectivity reflects the central role of the selection principle in determining the physical content of the theory. **Least connected axioms:** A_V and A_VI each have out-degree 1, making them the most targeted axioms in the dependency graph — each underwrites exactly one prediction directly. However, this does not make them less essential: P_6 (measurement consistency) and P_7 (finite particle spectrum) are each underwritten by exactly one axiom (A_V and A_VI respectively), making those axioms uniquely

responsible for those predictions. **Most supported prediction:** P_5 (Born rule) has the highest in-degree (3 axioms: A_I, A_III, A_IV all contribute to it), reflecting the derivational complexity of the Born rule, which requires phase primacy, admissibility, and evolution simultaneously. **Most uniquely supported prediction:** P_6 (measurement consistency) is underwritten by A_V alone among direct dependencies, making Axiom V the unique guardian of Phase Theory's measurement theory. The graph reveals Phase Theory's axiomatic architecture as a genuinely networked structure, not a linear deductive chain: each prediction has multiple supporting axioms, and each axiom supports multiple predictions, creating a web of mutually reinforcing dependencies that gives the theory its structural robustness.

29. Comparison with Other Foundational Theories

We place Phase Theory's axiomatic structure in context by comparing it with the leading foundational frameworks of modern physics. For each theory, we identify the number of ontological primitives, the typical number of axioms in a modern formalization, the degree of redundancy documented in the literature, and the level of ontological economy achieved. This comparison reveals Phase Theory's position as the most axiomatically economical of all competing frameworks.

Theory	# Ontological Primitives	# Axioms (modern formal.)	Known Redundancy	Ontological Economy
Classical Mechanics (Newton)	4 (space, time, mass, force)	6-8	High: Law 1 is special case of Law 2	Very Low
Special Relativity (Einstein)	2 (spacetime, c)	2-3	Low: postulates nearly	Moderate

Theory	# Ontological Primitives	# Axioms (modern formal.)	Known Redundancy	Ontological Economy
1905)			independent	
General Relativity	3 (manifold, metric, stress-energy)	4-5	Moderate: Bianchi identities create dependence among field equations	Moderate
Quantum Mechanics (von Neumann)	5-6 (Hilbert space, observables, unitarity, collapse, Born rule, state space)	5-6	High: Born rule independence debated; collapse postulate contested	Low
Quantum Field Theory (Standard Model)	Effectively uncountable (each field, coupling, gauge group factor)	Not axiomatized	Not formally analyzed; very high expected	Very Low
String Theory	Very high (strings, extra dims, D-branes, flux vacua, dualities)	Not fully axiomatized	Landscape of 10^{500} vacua suggests very high redundancy	Very Low
Loop Quantum Gravity	3-4 (spin networks/foams, Barbero-Immirzi parameter, Hamiltonian constraint)	4-6	Moderate: Hamiltonian constraint implies some equations; constraint algebra debate	Low
Causal Set Theory	2 (events, causal order)	3-4	Low: postulates nearly independent	High
Phase	1 (Φ)	6	None	Maximal

Theory	# Ontological Primitives	# Axioms (modern formal.)	Known Redundanc y	Ontological Economy
Theory			(proved in this paper)	

Table 4. Comparison of foundational theories by ontological economy and axiomatic redundancy.

The pattern in Table 4 is striking. Theories with more primitives generally have more axioms and more redundancy. The Standard Model, despite being the most empirically accurate physical theory ever constructed, is axiomatically the least disciplined: it is not formally axiomatized, its primitive count grows with each new observed particle, and the redundancy among its many parameters and symmetry groups has not been systematically analyzed. String theory improves on this in some respects (the landscape of string vacua is generated by relatively simple rules) but dramatically worsens the primitive count by introducing extra dimensions, D-branes, and a vast landscape of possible compactifications. Loop quantum gravity achieves moderate ontological economy but retains the Barbero-Immirzi parameter as an unexplained primitive, and its Hamiltonian constraint exhibits known redundancy.

Among theories with genuine ontological reduction ambitions, causal set theory [26] comes closest to Phase Theory's economy: it posits just two primitives (discrete events and causal order) and achieves high ontological economy with few axioms. However, causal set theory retains fundamental discreteness as a primitive — precisely what Phase Theory derives as an emergent feature (Paper 3). This distinction places Phase Theory at a strictly higher level of ontological reduction than causal set theory, even though both theories share the ambition of deriving spacetime from more fundamental structures.

Classical mechanics and Newton's three laws deserve special comment in this comparison [11]. The first law (inertia: a body in uniform motion remains so unless acted upon by a net force) is derivable from the second law ($F = ma$, Eq. 15) by substituting $F = 0$, which gives $a = 0$, i.e., constant velocity — the content of the first law. This redundancy has been recognized for over a century but is typically preserved in presentations of classical mechanics because the first law has important conceptual content (it establishes the existence of inertial frames) that would be lost by treating it purely as a corollary. Phase Theory's minimality proof shows that Phase Theory is free of even this type of "pedagogically motivated redundancy": each of the six axioms contributes logically independent content that cannot be derived from the others, even partially.

30. The Ontological Economy Principle

The minimality analysis of this paper is a formal realization of a deeper principle: the **Principle of Ontological Economy**, which holds that a foundational physical theory should minimize the number and complexity of ontological primitives required to recover all physical predictions. This principle has a distinguished philosophical history, from Occam's Razor through Mach's principle to modern physicalism and the anti-metaphysical stance of logical positivism. In its original medieval formulation, Occam's Razor counsels that "entities should not be multiplied beyond necessity" — a slogan that captures the economy principle in its most general form. Mach applied this principle to mechanics: he sought to eliminate absolute space and absolute time as primitives by replacing them with relational facts about the distribution of matter [20]. Einstein's special relativity achieved a partial realization: the aether (an ontological primitive of nineteenth-century physics) was eliminated, and the speed of light was treated as a structural feature of spacetime rather than a property of a medium. General relativity went further by eliminating gravitational force as a primitive, replacing it

with spacetime curvature — a single structural feature of a single primitive (the metric) that accounts for the previously separate concepts of inertia and gravity.

The minimality theorem proved in Section 24 is a formal realization of the economy principle in the context of Phase Theory: it proves not merely that Phase Theory is economical (which could be established by inspection) but that it is *maximally* economical within its stated scope. The proof shows that S is the smallest axiom set that still generates P — no subset of S suffices. Combined with the single-primitive structure (Axiom I provides the only ontological primitive), Phase Theory achieves economy at both levels simultaneously: the minimum number of primitives (one) and the minimum number of axioms required to generate the theory's physical content (six, as proved in this paper).

An important distinction must be drawn between ontological economy and simplicity of the axiom set. A set of axioms can be *syntactically simple* — short, easy to state — while postulating many ontological primitives. String theory's axioms can be stated relatively briefly (one-dimensional objects, Nambu-Goto action, quantum mechanics), yet the resulting framework posits enormously many primitives through the landscape of vacua. Conversely, a set of axioms can be *syntactically complex* — lengthy, technically demanding — while postulating few primitives. Phase Theory's six axioms are not trivially simple to state (Axiom III in particular has significant technical content), but they postulate a single primitive. Ontological economy is a property of the latter kind: it concerns the inventory of reality committed to by the theory, not the verbal economy of the theory's axiomatic presentation. Phase Theory is economical at the ontological level; it may appear complex at the syntactic level, but this apparent complexity is the result of specifying a rich structure from a minimal ontological base.

Is ontological economy a virtue in itself, or merely instrumentally valuable? We argue it is both. Instrumentally, ontological economy is valuable because theories with fewer primitives are more constrained and therefore more falsifiable: each primitive can be varied independently, and fewer primitives means fewer dimensions of possible variation, yielding sharper predictions. Phase Theory's single-primitive structure makes its predictions dramatically more constrained than the Standard Model's multi-parameter predictions: a finite particle spectrum bounded by approximately 25 ± 10 species cannot be adjusted by tuning free parameters, because there are no free parameters at the axiomatic level. Each prediction is a fixed consequence of the single-primitive, six-axiom framework. This epistemic virtue is directly tied to ontological economy. As an aesthetic virtue, ontological economy reflects a deep structural feature of successful physical theories: the most revolutionary advances in physics (special relativity, quantum mechanics, general relativity) have each eliminated primitives rather than adding them. Phase Theory, by eliminating all primitives except one, represents the logical endpoint of this historical trend.

31. Axiom Strengthening: What Happens If We Strengthen Rather Than Weaken?

The minimality analysis establishes that no axiom can be weakened without loss of physical predictions. The complementary question — whether any axiom could be strengthened without gain — is equally important for confirming the calibration of the axiom set. If an axiom could be strengthened (replaced by a logically stronger statement) without changing the theory's predictions, then the current axiom is unnecessarily weak and fails to express the full extent of Phase Theory's commitments. We examine each axiom for possible strengthening and show that all strengthenings are either empirically destructive or logically equivalent to the current statement.

Strengthening Axiom II (Continuity): The current axiom requires C^1 continuity. A natural strengthening is to require C^∞ (smooth) or even real-analytic continuity of Φ everywhere, with no admissible singularities. In a C^∞ theory without singularities, Φ is a globally smooth field. This strengthening is immediately destructive: the admissible singularities of Axiom II are the topological defects of Φ that correspond to particle-like structures in Paper 7. A globally smooth Φ with no singularities has no particle-like structures: the particle spectrum is empty, and P_7 fails trivially (zero species instead of approximately 25 ± 10). The strengthening is therefore empirically inadmissible. Alternatively, one could strengthen Axiom II to C^∞ on $D \setminus S$ while retaining admissible singularities. This is logically possible, but it introduces mathematical complications: the evolution equations for Φ are derived from $A[\Phi]$ and are determined by $A[\Phi]$'s form, not by the regularity class of Φ . Making Φ smoother than $A[\Phi]$ requires would over-constrain the solution space in a way not warranted by the physics. The C^1 specification is the natural regularity class for the theory as formulated.

Strengthening Axiom III (Admissibility): The current axiom permits all extrema of $A[\Phi]$ — both minima and saddle points. A natural strengthening is to restrict admissible configurations to strict local minima only (no saddle points, no maxima). This strengthening eliminates saddle points from the admissible configuration space. However, saddle points of $A[\Phi]$ play an essential physical role: they correspond to unstable but physically important configurations — resonances (which appear as poles in scattering amplitudes), tunneling configurations (instantons, which mediate quantum tunneling between degenerate vacua), and virtual processes (off-shell configurations that contribute to quantum amplitudes in the path integral formulation). Eliminating saddle points would destroy the scattering amplitude computations of Paper 7 and the tunneling phenomena implicit in the thermodynamics of Paper 8. The strengthening is empirically destructive.

The current formulation — admissibility by extremization — is precisely the level of generality required.

Strengthening Axiom IV (Evolution): The current axiom requires that evolution be an ordered sequence of admissible transformations, without specifying whether this sequence is deterministic. A natural strengthening is to require strict determinism: at each stage, there is a unique admissible successor configuration. This strengthening would make Phase Theory a deterministic theory. However, Phase Theory's derivation of the Born rule (Paper 5) requires quantum mechanical probabilities, which are derivable in Phase Theory precisely because the evolution is not strictly deterministic at the level of individual configurations: multiple admissible successors exist, and the probability of each is determined by the $A[\Phi]$ -induced measure. A strictly deterministic Phase Theory (one satisfying the strengthened Axiom IV) would have no quantum mechanical probability and would fail to reproduce the probabilistic predictions of quantum mechanics. The strengthening is empirically destructive.

Strengthening Axiom V (Observability): The current axiom permits three types of observables: invariants, extrema, and topological features. A natural strengthening is to restrict to invariants only, excluding extrema and topological features. Extrema of $A[\Phi]$ include on-shell particle properties (masses, coupling strengths) that are directly measured in experiments. Topological features include winding numbers and Chern numbers that classify particle species (Paper 7). Excluding extrema would eliminate all on-shell observables; excluding topological features would eliminate all topological quantum numbers. Both exclusions destroy large portions of Phase Theory's empirical contact with experiment. The strengthening is empirically destructive. The tripartite classification of Axiom V is the minimal complete characterization of physical observables, and any restriction of it is excessive. We conclude that no axiom of Phase Theory can be fruitfully

strengthened: all strengthenings are either empirically destructive or physically unmotivated. The axiom set is calibrated not only from below (no weakening without loss) but also from above (no strengthening without destructive over-constraint).

32. Axiom Decomposition: Can Any Axiom Be Split?

Axiom decomposition is the third type of potential modification of an axiom set, alongside elimination (Section 24) and strengthening/weakening (Section 31). An axiom A_k is **decomposable** if it is a conjunction of two logically independent statements, say $A_k = B \wedge C$, where B and C are independent in the sense that neither implies the other. A decomposable axiom should be listed as two axioms in a maximally refined presentation. We examine each of the six axioms for decomposability.

Axiom I decomposition analysis: Axiom I reads "There exists Φ (a global phase substrate) such that every physical entity is an emergent structure of Φ , and no physical entity exists independently of Φ ." This can be formally analyzed as a conjunction of: (Ia) $\exists\Phi$ (the global phase substrate exists); and (Ib) $\forall$ physical entity E : E is an emergent structure of Φ (every entity is phase-theoretic). Could (Ia) be logically independent of (Ib)? Note that (Ia) alone asserts only the existence of Φ as some entity in the theory; it does not specify Φ 's relationship to other entities. (Ib) alone presupposes the existence of Φ (one cannot say "every entity is a structure of Φ " without Φ existing) but asserts additionally that the relationship between Φ and everything else is one of grounding or emergence. The conjunction $(Ia) \wedge (Ib)$ is standard existence-plus-comprehensiveness, and attempting to split it would produce (Ia) as a trivially weak statement (Φ exists as an ingredient of the theory, without specifying its role) and (Ib) as a statement that presupposes (Ia). The split is not a genuine logical decomposition because the two components are

not logically independent — (Ib) semantically presupposes (Ia). Axiom I is irreducibly conjunctive in this sense: its two components cannot be asserted independently. No genuine decomposition is achieved.

Axiom III decomposition analysis: Axiom III reads "Not all phase configurations are physically realizable; realizable configurations are those that extremize $A[\Phi]$ subject to topological and boundary constraints." This could be split into: (IIIa) Not all configurations are realizable (a selection principle exists); and (IIIb) Realizable configurations extremize $A[\Phi]$ (the selection principle is variational). We examine whether (IIIa) and (IIIb) are logically independent. (IIIb) logically implies (IIIa): if realizable configurations are specifically those extremizing $A[\Phi]$, then not all configurations are realizable (in particular, non-extremal configurations are not). Therefore (IIIa) is logically entailed by (IIIb) and is redundant given (IIIb). No genuine logical decomposition is achieved: the apparent conjunction reduces to (IIIb) alone, with (IIIa) as its consequence. Axiom III as stated is therefore equivalent to its stronger component (IIIb) alone. The current formulation includes (IIIa) for conceptual clarity — it emphasizes that the theory involves selection — but does not add independent logical content. This is a case where the axiom contains a rhetorically motivated redundancy that is not a logical redundancy: the axiom is not a conjunction of two independent statements but a single statement with a helpful preamble. No decomposition is required.

Axiom V decomposition analysis: Axiom V identifies three categories of observables: (Va) gauge-invariant invariants, (Vb) extrema of $A[\Phi]$, and (Vc) topological features of Φ . Could Axiom V be decomposed into three axioms, one for each category? We analyze whether these three categories are logically independent within the Phase Theory framework. (Va) and (Vc) are independent: topological features (such as winding numbers) are gauge-invariant but are not in general extrema of $A[\Phi]$; and invariants of $A[\Phi]$'s

symmetry group are not in general topological. (Vb) and (Va) overlap: on-shell extrema of $A[\Phi]$ include gauge-invariant quantities, but not all invariants are extrema. The three categories are thus partially overlapping and partially independent — they form a natural unified class (gauge-invariant and physically meaningful features of Φ) that is most efficiently expressed as a single axiom. Splitting into three axioms would create three axioms where one suffices, increasing the axiom count without increasing the content. The current formulation is the most compact expression of the observability constraint. No beneficial decomposition is achieved. We conclude that none of the six axioms of Phase Theory admits a non-trivial logical decomposition. The axiom count of six is exact: it is neither an undercount (no axiom conceals a hidden conjunction of independent assertions) nor an overcount (no axiom is redundant).

33. Objections and Replies

We now address systematically the principal objections that have been raised or could be raised against the minimality analysis and the axiomatic structure of Phase Theory.

Objection 1: "Closure (Axiom VI) is a philosophical preference, not a physical axiom." This objection holds that Axiom VI expresses a methodological preference for ontological parsimony — a scientific virtue, perhaps, but not a physical postulate with empirical consequences.

Reply: Axiom VI is physically substantive because it generates a falsifiable prediction: the particle spectrum is finite and bounded (Paper 7 [5]). If new particles are discovered outside the predicted range of approximately 25 ± 10 species and if those particles provably cannot be derived as topological structures of Φ , Phase Theory is falsified. A philosophical preference would not generate testable predictions with determinate failure conditions. Furthermore, Axiom VI determines the research methodology of Phase

Theory in empirically consequential ways: in the face of a new observation (say, a new particle or interaction), Phase Theory mandates a search for a phase-theoretic derivation, while a theory without Axiom VI permits parameter addition. These two responses lead to genuinely different theoretical and experimental programs. The difference is empirically discernible: Phase Theory's response to anomalies is derivation, which generates new predictions and constrains the theory further; parameter addition generates new predictions only after fixing the new parameters. Axiom VI is therefore physically substantive in its methodological implications as well as its direct predictions.

Objection 2: "Observability (Axiom V) is subjective — the choice of 'gauge-invariant' features is conventional."

Reply: Axiom V's criterion for observability is not subjective. A quantity $F[\Phi]$ is an observable in Phase Theory if and only if F satisfies a precise mathematical condition: it is gauge-invariant (unchanged by all admissible gauge transformations of Φ), or it is a value of $A[\Phi]$ at an extremal configuration, or it is a topological invariant of Φ . Each of these conditions is a precisely stated mathematical criterion with no element of subjectivity. The choice of what constitutes a "gauge transformation" is determined by the symmetry structure of $A[\Phi]$, which is itself constrained by the predictions of Papers 7–10. What may appear conventional is the choice of which symmetries are "gauge" (conventional) versus "global" (physical): but this distinction, while requiring specification as part of the model-level input to Phase Theory, is not subjective once the model is specified — it is determined by the mathematical structure of $A[\Phi]$. Axiom V's criterion is therefore objective given the specification of $A[\Phi]$, and $A[\Phi]$ is constrained by empirical predictions to a finite class of functional forms.

Objection 3: "Admissibility (Axiom III) is arbitrary — why should $A[\Phi]$ have any particular form, and why should extremization rather than, say, minimization or some other principle be the selection criterion?"

Reply: Axiom III does not fix the form of $A[\Phi]$; it asserts that a global selection functional of the extremization type exists. The choice of extremization (rather than, say, minimization alone) is not arbitrary but motivated: as noted in Section 31, strict minimization would eliminate saddle points and destroy the scattering theory of Paper 7. The extremization principle is the minimal selection criterion that preserves both stable configurations (minima) and physically important unstable configurations (saddle points). More generally, the extremization principle — Euler's principle of stationary action — has an extraordinarily successful track record in foundational physics: classical mechanics, electromagnetism, general relativity, and quantum field theory all employ it. Its appearance in Phase Theory is not arbitrary but reflects the deepest pattern of physical theorizing. The specific form of $A[\Phi]$ is not arbitrary either: it is constrained by the requirement that Phase Theory recover the known particle spectrum, thermodynamics, and spacetime geometry. The set of $A[\Phi]$ consistent with all of Phase Theory's predictions is not the set of all functionals but a restricted family whose characterization is an active research program.

Objection 4: "Continuity (Axiom II) conflicts with quantum gravity, which suggests a minimum length (Planck length) at which the continuity of spacetime breaks down."

Reply: This objection conflates two levels of description. Axiom II asserts the continuity of the phase substrate Φ , not the continuity of emergent spacetime. Spacetime emerges from Φ in Paper 10; it is a secondary structure, not the fundamental entity. A minimum effective length scale in emergent spacetime is entirely compatible with a continuous Φ : the minimum length arises as an emergent feature of the admissibility structure of $A[\Phi]$, reflecting the fact

that phase configurations below a certain scale are topologically inadmissible, not that Φ itself is discrete. Phase Theory therefore predicts a minimum length (consistent with quantum gravity phenomenology) as a derived rather than a fundamental feature. Axiom II, by forbidding fundamental discreteness, makes this derivation non-trivial and ensures that the minimum length is a prediction rather than a presupposition. The objection, properly understood, is not a criticism of Axiom II but a confirmation of one of its predictions: that minimum length is emergent, not fundamental.

Objection 5: "The independence proofs of this paper are not fully rigorous — the 'models' are physical theories, not formal mathematical structures, and the notion of 'satisfaction' is informal."

Reply: The independence proofs of Sections 6, 9, 12, 15, 18, and 21 each construct explicit models satisfying all but the target axiom, with sufficient precision to establish the relevant independence claims within the physical framework. The models (dualistic theory, lattice phase theory, unconstrained phase theory, static phase landscape, unrestricted observable theory, open-ontology phase theory) are defined precisely enough to determine which axioms they satisfy and which they violate. A fully formalized treatment within a specific mathematical framework (category theory, topos theory, or a formal language for physical theories following Suppes [see 9]) would strengthen the results but would not change their content: the distinguishing models constructed here are already mathematically well-defined physical theories, and their properties are determined by their mathematical specifications. The formalization program is valuable and would benefit from being pursued as a companion mathematical paper; but the informal-formal gap in the current proofs is one of presentation rather than substance.

Objection 6: "Six axioms is still too many — a truly fundamental theory should have even fewer, perhaps three or four."

Reply: The structural classification of Section 27 shows that the six axioms occupy five distinct functional roles (ontological $\times 2$, regularity $\times 1$, selection $\times 1$, dynamical $\times 1$, empirical $\times 1$). A theory missing any one of these roles suffers a corresponding type of theoretical failure. Five axioms would require one functional role to go unrepresented; the minimality theorem proves that any five-axiom subset of S fails to recover at least one prediction in P. Three or four axioms would miss even more functional roles and would fail to recover multiple predictions. The question is not how few axioms one would prefer but how few axioms are sufficient to generate the theory's physical content. The proof shows that the answer is exactly six. The intuition that "fewer is better" is sound as a guiding principle but must be disciplined by the formal result: six is not six because we stopped counting, but because removing any one of the six causes a specific, identifiable physical failure.

34. The Relationship Between Axioms VI and I

We examine in detail the logical and conceptual relationship between Axiom I (Phase Primacy) and Axiom VI (Closure), since both are ontological axioms and their relationship raises the most natural suspicion of redundancy. The *prima facie* argument for redundancy runs as follows: if Axiom I asserts that Φ is the fundamental substrate from which everything emerges, then surely this implies that no other primitives exist — which is precisely what Axiom VI asserts. If this argument were correct, Axiom VI would be a corollary of Axiom I, and the axiom count could be reduced to five. We show that this argument fails at a crucial juncture.

The key distinction is temporal or developmental: **Axiom I is a claim about the present structure of the theory's axioms**, while **Axiom VI is a constraint on the theory's future development**. Axiom I says: in the current axiom set, Φ is the unique primitive. Axiom VI says: in all future

extensions of the axiom set, no new primitive may be introduced. These are categorically different types of claim. Axiom I is a synchronic claim (about the current state of the theory); Axiom VI is a diachronic or regulative claim (about how the theory is to be developed over time). A theory can satisfy Axiom I (Φ is currently the sole primitive in its axiom set) while violating Axiom VI (new primitives are introduced when empirical anomalies arise): this is precisely the model M_6 of Section 21, which satisfies Axioms I-V but violates Axiom VI.

The logical structure illuminates the relationship further. Axiom I is an existential claim: $\exists \Phi: \forall E (E \text{ is physical} \rightarrow E \text{ is a feature of } \Phi)$. Axiom VI is a universal prohibition: $\forall$ extensions of the theory (no new primitive is introduced). Neither of these claims implies the other. Axiom I does not imply Axiom VI because Axiom I does not rule out a theorist subsequently introducing a new primitive χ and revising the theory to include χ as a co-primitive alongside Φ . After such a revision, Axiom I might still hold in a weakened sense (Φ is primary), but Axiom VI is violated. Axiom VI does not imply Axiom I because Axiom VI prohibits additions to the primitive set but says nothing about what is currently in the primitive set: a theory could comply with Axiom VI (no additions ever) while starting with two primitives (Φ and something else). Axiom I specifies what the current primitive is (Φ and only Φ); Axiom VI ensures that this specification remains stable under future development.

An analogy illuminates the relationship: Axiom I is the **existence claim** and Axiom VI is the **uniqueness claim**. In mathematics, existence and uniqueness are separate statements that are jointly expressed in the form "there exists a unique X such that..." ($\exists! X: P(X)$). The existence component asserts that X exists; the uniqueness component asserts that no other object satisfies the same conditions. In Phase Theory, Axiom I plays the role of the existence claim (Φ exists and is fundamental) and Axiom VI plays the role of

the uniqueness guarantee maintained over time (no other primitive will ever be recognized alongside Φ). Together they say: " Φ is and will remain the unique ontological primitive of Phase Theory." Separately, each makes a weaker claim. The joint assertion of both is logically stronger than either individually, confirming their independent necessity.

35. Implications for Falsifiability

The minimality theorem has direct and important consequences for the falsifiability of Phase Theory. Because every axiom is load-bearing for at least one physical prediction, every axiom is potentially falsifiable through its downstream predictions. This means that Phase Theory is not merely falsifiable as a whole — if any prediction fails, the theory is falsified — but is **diagnostically falsifiable**: a failed prediction pinpoints which axiom is responsible, providing precise information about which fundamental commitment of the theory has been empirically violated. This diagnostic precision is a direct consequence of the non-redundancy and minimality of the axiom set, and is a significant epistemic virtue that distinguishes Phase Theory from theories with redundant or entangled axiom sets.

Axiom I (Phase Primacy) is falsifiable through any observation that demonstrably cannot be accounted for by any phase-theoretic derivation and requires a new ontological primitive. The operative word is "demonstrably": Phase Theory is not falsified by a failed derivation attempt, only by a proof that no phase-theoretic derivation is possible for the observed phenomenon. Such a proof would constitute a direct refutation of Axiom I's claim that everything is a feature of Φ . While such a proof would be extraordinarily difficult to establish, the logical structure of the falsification condition is clear: it requires either a no-go theorem within Phase Theory (analogous to Bell's theorem for hidden variables) or an empirical observation whose phase-theoretic description would require logical contradiction.

Axiom II (Continuity) is falsifiable through the observation of fundamental discreteness at a scale not derivable from $A[\Phi]$. Concretely, if experiments were to reveal a minimum length structure with no corresponding prediction from Phase Theory's admissibility functional — a discrete structure that is fundamental rather than emergent — Axiom II would be violated. Such an observation would likely arise in ultra-high-energy scattering experiments probing distances below the Planck scale, or in precision tests of spacetime continuity in gravitational wave astronomy. Axiom III (Admissibility) is falsifiable through the observation of particle species outside the predicted range of approximately 25 ± 10 or through the discovery of a thermodynamic process that cannot be accounted for by the $A[\Phi]$ -derived equilibrium structure. The finite particle spectrum prediction is particularly sharp: the observation of a new particle requires either a phase-theoretic derivation of that particle from $A[\Phi]$ (acceptable within Phase Theory) or an acknowledgment that the particle lies outside the admissible spectrum (falsifying Axiom III's prediction of a bounded spectrum).

Axiom IV (Evolution) is falsifiable through the observation of acausal phenomena — events that violate the ordering relation $<$ on admissible phase transformations — or through the demonstration that the universe is time-symmetric at the cosmological level, contradicting the arrow-of-time prediction P_8 . Axiom V (Observability) is falsifiable through the discovery of an experimentally confirmed physical observable that cannot be expressed as a gauge-invariant invariant, extremum, or topological feature of Φ . Axiom VI (Closure) is falsifiable through the discovery of a phenomenon that provably requires a new ontological primitive and cannot be derived from Φ . The provability condition is essential: Axiom VI is falsified only when derivation from Φ is demonstrated to be impossible, not merely difficult.

The diagnostic falsifiability of Phase Theory is a major epistemic advantage over frameworks like string theory, which are notoriously difficult to falsify

precisely because their parameter space is vast (the landscape of 10^{500} vacua) and their primitive set is open-ended (new D-branes and fluxes can be introduced as needed). In string theory, a failed prediction does not identify which commitment of the theory is responsible; it merely motivates exploring a different part of the landscape. Phase Theory, by contrast, is structured so that each failed prediction points to a specific axiom. This makes the theory scientifically tractable in the classical falsificationist sense: it is not merely in principle falsifiable but practically falsifiable in the sense that falsification has a precise and diagnostically informative structure.

36. Connections to Gödel Completeness and Consistency

The independence analysis of this paper has a formal kinship with independence proofs in mathematical logic, and it is illuminating to examine this kinship carefully, both to clarify the logical status of the paper's results and to identify their limits. The most celebrated independence results in mathematics are Gödel's incompleteness theorems [23] and the independence of the Continuum Hypothesis from ZFC [24]. These results establish that certain statements — the consistency of arithmetic, the Continuum Hypothesis — cannot be proved or disproved from specified axiom sets. They are proved by model-theoretic arguments: constructing models of the axioms in which the statement holds (relative consistency) and models in which it fails (independence).

The independence proofs of Sections 6–21 follow the same structural pattern: for each axiom A_k , we construct a model M_k satisfying all other axioms but violating A_k . This is the model-theoretic technique of establishing independence. The analogy is genuine and substantive: the logical structure of the independence argument is identical. The constructions of $M_1, \dots, M_6$ are physical-theoretic analogues of Cohen's forcing construction for the

independence of the Continuum Hypothesis [24], or Gödel's constructible universe for its relative consistency. The conceptual framework is the same: to show that a statement is independent of an axiom set, exhibit a model of the set in which the statement is false.

However, important disanalogies must be acknowledged. Gödel's incompleteness theorems apply to formal systems strong enough to express elementary arithmetic — specifically, to systems that are consistent, recursively axiomatizable, and sufficiently expressive. They establish not merely the independence of specific statements but the existence of statements independent of *any* consistent extension of the axiom set. Phase Theory is not a formal system in this sense: it is a physical theory, and its axioms are claims about physical reality rather than axioms of a formal calculus. The notion of "satisfaction" employed in Sections 6–21 is physical rather than formal: a model "satisfies" an axiom if the model's physical structure is consistent with the statement of that axiom. This is an informal rather than a formal notion of satisfaction, and it lacks the complete precision of formal model theory.

Gödel's completeness theorem establishes that, for first-order theories, semantic independence (existence of a model violating the statement) implies syntactic independence (no formal proof of the statement). For Phase Theory, a fully syntactic independence result would require: (i) formalizing Phase Theory within a specific formal language (first-order, second-order, or topos-theoretic); (ii) specifying deductive rules for that language; (iii) proving that no formal proof of A_k exists from $\{A_j : j \neq k\}$ within the specified system. This program would constitute a rigorous metamathematical treatment of Phase Theory and is an important direction for future foundational work. The proofs of the present paper establish the relevant independence results at the level of precision appropriate for a physical theory: the distinguishing models are specified precisely enough to determine their physical properties, and the

logical arguments are conducted at a level of rigor comparable to standard mathematical physics.

The relationship with Gödel's incompleteness is also worth examining. Incompleteness would mean that there exist physically meaningful statements about Phase Theory that are neither provable nor disprovable from the six axioms. This is entirely possible and would not be surprising: Gödel's theorem guarantees the existence of such statements in any sufficiently rich formal system. The minimality result of this paper does not preclude incompleteness: it establishes that the six axioms form a minimal sufficient set for the specific predictions P , not that they form a complete deductive system for all physically meaningful statements. The investigation of which physical statements about Φ are undecidable from the six axioms — and whether those statements are physically significant — is an open and important question for Phase Theory's foundational development.

37. Implications for Theory Extension

The minimality theorem has profound implications for how Phase Theory can legitimately be extended to new physical domains. Since the axiom set is minimal with respect to the core predictions P , and since Axiom VI prohibits the introduction of new primitives, the only legitimate mode of extending Phase Theory is **derivation**: proving new theorems from the existing axioms. This severely disciplines the research program and distinguishes Phase Theory from effective field theory frameworks that can accommodate new phenomena by adding new operators or parameters.

The most pressing extension challenges for Phase Theory involve phenomena not fully accounted for in the current series of papers: dark matter, dark energy, the cosmological constant problem, the hierarchy problem, and the black hole information paradox. Axiom VI mandates a specific response to each of these: find a phase-theoretic derivation or acknowledge falsification.

There is no third option of parameter addition. For **dark matter**, Phase Theory must derive the observed dark matter phenomenology (galactic rotation curves, gravitational lensing, structure formation) from the phase configuration of Φ . This would likely proceed through a class of phase configurations — topological defects or coherent phase structures — that interact gravitationally but not electromagnetically. The prediction of specific properties (mass, cross-section, distribution) of the "dark phase" structure would then be a derived consequence, not a postulated one.

For **dark energy**, Phase Theory must derive the observed accelerated expansion of the universe — attributed in the standard cosmological model to a cosmological constant Λ or a dynamical dark energy field — from the global topology of Φ . The cosmological constant problem (why Λ is approximately 120 orders of magnitude smaller than naive quantum field theory estimates) would be reframed as a question about the admissibility conditions on the global phase topology: which topological sectors of Φ give rise to a small effective cosmological constant, and what determines which sector is realized? This is a specific, computationally tractable question within Phase Theory's framework, unlike in quantum field theory where the cosmological constant problem is a tuning problem with no structural resolution.

The minimality result also determines how Phase Theory responds to future experimental discoveries. Suppose a new particle is discovered beyond the predicted range of approximately 25 ± 10 species. There are three possible responses within Phase Theory: (i) the new particle is derivable from $A[\Phi]$ as a topological structure whose topological class was incorrectly excluded in the counting of Paper 7 — in which case the bound is revised but the theory is not falsified; (ii) the new particle is not derivable from any admissible Φ -configuration — in which case Phase Theory is falsified by the new particle; (iii) the new particle can be derived only by adjusting the admissibility functional $A[\Phi]$ — in which case the adjusted $A[\Phi]$ represents a refinement of

the Phase Theory model, remaining within the axiomatic framework. Note that response (iii) is legitimate precisely because Axiom III specifies the existence of $A[\Phi]$ without fully determining its form: different forms of $A[\Phi]$ represent different Phase Theory models, all consistent with the axioms. This model-level flexibility is the correct and necessary degree of flexibility for a foundational theory: the axioms are fixed; the model is refined by empirical input. Response (ii) is the genuine falsification condition, and its possibility is what makes Phase Theory scientifically tractable.

38. The Role of Each Axiom in Downstream Papers

We provide a comprehensive mapping of each axiom to the downstream papers of the Phase Theory series that depend upon it, distinguishing between essential dependence (the paper's central results cannot be stated or proved without the axiom) and supporting dependence (the axiom plays a role in some subsidiary results or background assumptions of the paper, though not in its central theorem). This mapping serves as a reference for readers of the Phase Theory series and makes explicit the foundational architecture of the entire program.

Axiom	Essential For (Central Results)	Supporting Role In (Subsidiary Results)
A_I Phase Primacy	Papers 1 (foundational axioms), 2 (coherence and propagation), 3 (discreteness), 4 (cosmology), 5 (probability), 6 (measurement), 7 (particle spectrum), 8 (thermodynamics), 9 (gauge interactions), 10 (spacetime)	All papers universally — Axiom I is the presupposition of every derivation in the series

Axiom	Essential For (Central Results)	Supporting Role In (Subsidiary Results)
A_II Continuity	Paper 3 (discreteness as emergent topology of continuous Φ), Paper 10 (smooth emergent spacetime via $\nabla\Phi$)	Papers 5, 7, 9 (smoothness of phase probability measure; smooth gauge field derivation)
A_III Admissibility	Paper 5 (Born rule from $A[\Phi]$ -induced measure), Paper 7 (finite particle spectrum from admissible topological classes), Paper 8 (thermodynamic equilibrium from $A[\Phi]$ minimum)	Papers 4, 9, 10 (cosmological initial conditions; gauge coupling structure; form of emergent metric)
A_IV Evolution	Paper 4 (cosmological history and expansion), Paper 8 (arrow of time from ordered phase transformations)	Papers 5, 6, 7 (probability requires sequences; measurement involves sequential transitions; scattering requires in/out state ordering)
A_V Observability	Paper 6 (measurement theory: consistency of measurement outcomes as gauge-invariant events)	Papers 5, 7 (gauge-invariant probability outcomes; gauge-invariant identification of particle properties)
A_VI Closure	Paper 7 (finiteness of particle spectrum: no new species can be postulated as primitives)	All papers (the derivation methodology of every paper — proving results from Φ rather than postulating new primitives — presupposes Axiom VI)

Table 5. Role of each axiom across the Phase Theory paper series.

Table 5 reveals the layered architecture of Phase Theory's foundations. Axiom I (Phase Primacy) is the universal foundation: it is essential for every paper in

the series, as every derivation begins from the assumption that Φ is the unique primitive. Axiom VI (Closure) plays a parallel role at the methodological level: it is "essential" for Paper 7's specific quantitative prediction, but it underwrites the entire research program of Phase Theory by mandating derivation over postulation in all papers. These two universal axioms — one asserting the existence of the single primitive, the other guaranteeing its permanence — form the backbone of the theory. Axioms II, III, IV, and V are more specialized in their essential roles: each is required for a specific subset of the downstream papers, corresponding to their structural roles (regularity, selection, dynamics, empirical contact respectively). The distribution of essential roles across the six axioms is non-uniform but complete: every paper in the series depends essentially on some proper subset of the axioms, and every axiom is essentially required by some proper subset of the papers.

39. Future Directions: Can the Axiom Count Be Reduced Below Six?

Having proved that the axiom count of six is irreducible within Phase Theory's current formulation and scope, we now speculate carefully on whether a reformulation of Phase Theory might achieve fewer than six axioms while maintaining the same predictive power. We approach this question with appropriate rigor, examining the two most plausible candidates for axiom consolidation and showing that both fail to achieve genuine logical reduction.

Candidate Consolidation 1: Merging Axioms I and VI. As established in Section 34, Axioms I and VI are the two ontological axioms, and they jointly express the claim " Φ is and will remain the unique ontological primitive." Could this joint claim be expressed as a single axiom: Axiom I* — " Φ is the unique and permanent ontological primitive of Phase Theory"? The linguistic merger is straightforward. The logical question is whether the merged axiom

Axiom I* has the same deductive power as the conjunction of Axioms I and VI separately. In one sense, yes: Axiom I* asserts both the existence of Φ as the current sole primitive (the content of A_I) and the prohibition on future additional primitives (the content of A_VI). The merged statement is logically equivalent to the conjunction $A_I \wedge A_{VI}$ and therefore has the same deductive consequences. However, the merger creates a practical difficulty for the independence analysis: when analyzing the independence of Axiom I*, one must construct a single distinguishing model that violates both A_I and A_VI simultaneously, rather than separate models for each. This is less informative: the independence of A_I and A_VI from each other (which we proved in Section 26) is concealed by the merger. The merger is therefore a notational convenience that obscures the fine structure of the axiom set without achieving genuine logical reduction. The axiom count remains six in the relevant sense: six independent logical commitments, even if two of them are listed under a single axiom heading.

Candidate Consolidation 2: Merging Axioms III and IV. As the selection axiom (Axiom III) and the dynamical axiom (Axiom IV) both concern the temporal evolution of admissible phase configurations, one might attempt to merge them: Axiom III* — "Physical evolution is the ordered extremization of $A[\Phi]$, with realizable configurations being those that successively extremize $A[\Phi]$ along the ordered sequence." This merged statement combines the existence of $A[\Phi]$ (Axiom III's content) with the ordering of configurations (Axiom IV's content). Does the merger achieve genuine logical reduction? No: the distinguishing model M_4 (static phase landscape, Section 15) demonstrates that Axiom III's content — the existence of $A[\Phi]$ selecting admissible configurations — is logically independent of Axiom IV's content — the ordering of those configurations. M_4 is a theory with $A[\Phi]$ but no ordering: it satisfies Axiom III but violates Axiom IV. No merged Axiom III* can be satisfied by M_4 (since M_4 lacks ordering) while the original Axiom III is satisfied by M_4 (since it does have $A[\Phi]$). Therefore the merged Axiom III*

is strictly stronger than Axiom III alone: it rules out models that Axiom III permits (specifically, static admissibility classification theories like M_4). Since M_4 satisfies Axiom IV-stripped Axiom III but violates Axiom IV, the two axioms have logically distinct content that cannot be captured by a single merged statement without either losing content or strengthening one of the components. The merger fails.

We also consider whether a completely different formulation of Phase Theory — using a different set of primitive concepts, perhaps a category-theoretic or topos-theoretic formulation — might achieve the same physical content with fewer axioms. This possibility cannot be ruled out by the results of this paper, since the minimality theorem proves minimality relative to the specific axiom set S as formulated, not relative to all possible axiom sets for Phase Theory. A reformulation in a richer mathematical language might encode multiple current axioms into a single, more sophisticated axiom. For example, a formulation using the language of higher category theory might be able to express both the existence of $A[\Phi]$ and the ordering of configurations as a single categorical structure (for example, a $(\infty,1)$ -category of admissible phase configurations with morphisms corresponding to admissible transformations). Whether such a reformulation genuinely reduces the axiom count or merely encodes the same commitments in a different notation is a subtle question whose answer depends on what one counts as a single axiom in the new language. We leave this as an open problem for future foundational work. Within the current formulation, the axiom count of six is irreducible.

40. Conclusion

This paper has conducted a systematic and formal analysis of the axiomatic foundation of Phase Theory with respect to independence, minimality, and non-redundancy. The central result is the Minimality Theorem (Theorem 1, Section 23), proved case-by-case in Section 24: the axiom set $S = \{A_I, A_{II},$

$A_{III}, A_{IV}, A_V, A_{VI}$ of Phase Theory is minimal with respect to the set of core physical predictions $P = \{P_3, P_5, P_6, P_7, P_8, P_{10}\}$. Removing any single axiom causes the irreversible loss of at least one core prediction, and the specific prediction lost is identified for each axiom. The evidence for this result is assembled from multiple independent analyses: six independence proofs (Sections 6, 9, 12, 15, 18, 21) constructing explicit distinguishing models (M_1 through M_6); the formal minimality proof (Section 24) proceeding case by case; the non-redundancy argument (Section 25) deriving non-redundancy from minimality by contraposition and supplementing it with a direct pairwise argument; the complete pairwise independence analysis (Section 26) exhibiting 30 distinguishing models covering all 15 axiom pairs; the structural classification by functional role (Section 27) showing that the six axioms occupy five distinct functional classes; and the dependency graph analysis (Section 28) revealing the networked architecture of Phase Theory's axiomatic support.

The paper has also established a comprehensive set of negative results, each of which strengthens the minimality claim from a different angle. No axiom can be derived from the remaining five (non-redundancy, Section 25). No axiom can be weakened without losing predictions (calibration from below, Sections 10, 16, and the necessity analyses). No axiom can be strengthened without over-constraining the theory and destroying predictions (calibration from above, Section 31). No axiom is a concealed conjunction of two independent statements that should be listed separately (decomposition analysis, Section 32). The axiom count of six is not an artifact of the formulation: it represents six genuinely distinct and irreducible commitments of the theory, corresponding to six distinct functional roles (ontological existence, ontological closure, regularity, selection, dynamics, and empirical contact) that any complete foundational physical theory must discharge.

The structural comparison of Section 29 situates Phase Theory in the broader landscape of foundational physics and establishes its unique position: among all theories surveyed (classical mechanics, special relativity, general relativity, quantum mechanics, quantum field theory, string theory, loop quantum gravity, causal set theory), Phase Theory achieves maximal ontological economy — a single primitive — combined with zero known axiom redundancy, proved formally in this paper. This combination is not achieved by any other theory in the comparison set. The ontological economy principle (Section 30) provides the philosophical context: the minimality theorem is a formal realization of the principle that a foundational theory should minimize its ontological commitments to those strictly required for its physical content, and Phase Theory satisfies this principle at both the primitive level (one primitive, Φ) and the axiom level (six axioms, each indispensable).

The implications for the research program of Phase Theory are stringent and specific (Section 37): the minimality result, combined with Axiom VI, restricts the legitimate modes of theory extension to derivation alone. Dark matter, dark energy, the cosmological constant, and any other future empirical discovery must be accommodated through phase-theoretic derivations from the existing axiom set, or the theory is falsified. This restriction is not a limitation but a virtue: it makes Phase Theory dramatically more constrained and therefore more falsifiable than frameworks that accommodate new phenomena through parameter addition. The diagnostic falsifiability structure of the theory (Section 35) further sharpens this virtue: each axiom is individually falsifiable through its downstream predictions, so that a failed prediction precisely identifies which foundational commitment of the theory has been empirically violated. This epistemic precision is a direct consequence of the non-redundancy and minimality of the axiom set.

We conclude with the following assessment: Phase Theory, as axiomatized in Paper 1 and analyzed in this paper, represents — to the author's knowledge —

the most axiomatically disciplined and ontologically economical foundational framework for comprehensive physical theory currently in the literature. Its six axioms are not approximations, heuristics, or working assumptions; they are irreducible commitments, each carrying specific physical content and each falsifiable through specific experimental predictions. The axiomatic case for Phase Theory is, in this paper's conclusion, as strong as the empirical case developed across the companion papers of this series: both converge on the same foundational structure, and each strengthens the other. The phase substrate Φ is not merely a useful mathematical tool; it is, within the Phase Theory framework, the unique, irreducible, and axiomatically indispensable primitive of physical reality.

References

1. [1] Hanks, M. Phase Theory: A Unified Theory of Matter, Light, and Geometry. Preprint series, Paper 1. Phase Theory Publications (2026).
2. [2] Hanks, M. Phase Theory Paper 3: Emergence of Discreteness from the Continuous Phase Substrate. Preprint series, Paper 3. Phase Theory Publications (2026).
3. [3] Hanks, M. Phase Theory Paper 5: Probability Emergence and the Born Rule from Phase Admissibility. Preprint series, Paper 5. Phase Theory Publications (2026).
4. [4] Hanks, M. Phase Theory Paper 6: Measurement Theory Without the Collapse Postulate. Preprint series, Paper 6. Phase Theory Publications (2026).
5. [5] Hanks, M. Phase Theory Paper 7: The Finite Particle Spectrum and Topological Classification of Admissible Singularities. Preprint series, Paper 7. Phase Theory Publications (2026).
6. [6] Hanks, M. Phase Theory Paper 8: Thermodynamics and the Arrow of Time from Ordered Phase Evolution. Preprint series, Paper 8. Phase Theory Publications (2026).

7. [7] Hanks, M. Phase Theory Paper 10: Emergence of Spacetime Geometry from Phase Gradients. Preprint series, Paper 10. Phase Theory Publications (2026).
8. [8] Hilbert, D. Über die Grundlagen der Geometrie. Mathematische Annalen 46 (1895). [Revised and expanded as: Grundlagen der Geometrie. Teubner, Leipzig (1899).]
9. [9] Kolmogorov, A. N. Foundations of the Theory of Probability. 2nd English ed. Chelsea Publishing, New York (1956). [Original: Grundbegriffe der Wahrscheinlichkeitsrechnung. Springer, Berlin (1933).]
10. [10] Zermelo, E. Untersuchungen über die Grundlagen der Mengenlehre I. Mathematische Annalen 65, 261–281 (1908).
11. [11] Newton, I. Philosophiae Naturalis Principia Mathematica. Royal Society, London (1687). [Translation: Mathematical Principles of Natural Philosophy. University of California Press (1999).]
12. [12] Einstein, A. Zur Elektrodynamik bewegter Körper. Annalen der Physik 17, 891–921 (1905). [On the Electrodynamics of Moving Bodies.]
13. [13] Dirac, P. A. M. Lectures on Quantum Mechanics. Belfer Graduate School of Science, Yeshiva University, New York (1964).
14. [14] von Neumann, J. Mathematische Grundlagen der Quantenmechanik. Springer, Berlin (1932). [Mathematical Foundations of Quantum Mechanics. Princeton University Press (1955).]
15. [15] Penrose, R. Twistor Algebra. Journal of Mathematical Physics 8, 345–366 (1967).
16. [16] Rovelli, C. Loop Quantum Gravity. Living Reviews in Relativity 11, 5 (2008).
17. [17] Polchinski, J. String Theory. Vols. I and II. Cambridge University Press, Cambridge (1998).
18. [18] Weinberg, S. The Quantum Theory of Fields. Vols. I, II, III. Cambridge University Press, Cambridge (1995–2000).
19. [19] Wald, R. M. General Relativity. University of Chicago Press, Chicago (1984).

- 20.[20] Mach, E. Die Mechanik in ihrer Entwicklung historisch-kritisch dargestellt. Brockhaus, Leipzig (1883). [The Science of Mechanics. Open Court, Chicago (1902).]
- 21.[21] Wheeler, J. A. Information, Physics, Quantum: The Search for Links. In: Zurek, W. H. (ed.), Complexity, Entropy and the Physics of Information. Addison-Wesley, Redwood City, CA (1990), pp. 3-28.
- 22.[22] Goroff, M. H. and Sagnotti, A. Quantum gravity at two loops. Physics Letters B 160, 81-86 (1985).
- 23.[23] Gödel, K. Über formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme I. Monatshefte für Mathematik und Physik 38, 173-198 (1931). [On Formally Undecidable Propositions.]
- 24.[24] Cohen, P. J. Set Theory and the Continuum Hypothesis. W. A. Benjamin, New York (1966).
- 25.[25] Connes, A. Noncommutative Geometry. Academic Press, San Diego (1994).
- 26.[26] Bombelli, L., Lee, J., Meyer, D., and Sorkin, R. D. Space-time as a causal set. Physical Review Letters 59, 521-524 (1987).
- 27.[27] Isham, C. J. Canonical quantum gravity and the problem of time. In: Ibort, L. A. and Rodríguez, M. A. (eds.), Integrable Systems, Quantum Groups, and Quantum Field Theories. NATO Advanced Study Institutes Series C, vol. 409, 157-287. Kluwer, Dordrecht (1993).
- 28.[28] Jacobson, T. Thermodynamics of spacetime: The Einstein equation of state. Physical Review Letters 75, 1260-1263 (1995).
- 29.[29] Padmanabhan, T. Emergent gravity paradigm: Recent progress. Modern Physics Letters A 30, 1540007 (2015).
- 30.[30] Verlinde, E. On the origin of gravity and the laws of Newton. Journal of High Energy Physics 2011(4), 029 (2011).
- 31.[31] Suppes, P. Introduction to Logic. Van Nostrand, Princeton (1957).
- 32.[32] van Fraassen, B. C. The Scientific Image. Oxford University Press, Oxford (1980).

- 33.[33] Penrose, R. The Road to Reality: A Complete Guide to the Laws of the Universe. Jonathan Cape, London (2004).
- 34.[34] Dirac, P. A. M. The Principles of Quantum Mechanics. 4th ed. Oxford University Press, Oxford (1958).
- 35.[35] Henneaux, M. and Teitelboim, C. Quantization of Gauge Systems. Princeton University Press, Princeton (1992).
- 36.[36] Bourbaki, N. Éléments de Mathématique: Théorie des Ensembles. Hermann, Paris (1968).
- 37.[37] Baez, J. C. and Muniain, J. P. Gauge Fields, Knots and Gravity. World Scientific, Singapore (1994).
- 38.[38] Hawking, S. W. and Ellis, G. F. R. The Large Scale Structure of Space-Time. Cambridge University Press, Cambridge (1973).
- 39.[39] Sorkin, R. D. Forks in the road, on the way to quantum gravity. International Journal of Theoretical Physics 36, 2759–2781 (1997).
- 40.[40] Witten, E. Topological quantum field theory. Communications in Mathematical Physics 117, 353–386 (1988).